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List of Acronyms

BEM	Boundary Elements Method	PEM	Poroelastic material
EPWE	Extended Plane Wave Expansion	PML	Perfectly Matched Layer
FEM	Finite Elements Method	PWE	Plane Wave Expansion
GEP	Generalized Eigenvalue Problem	SAFE	Semi-analytical Finite Elements
GM	Global Method	SCM	Spectral Collocation Method
HR	Helmholtz Resonator	SRR	Split-Ring Resonator
JCA	Johnson-Champoux-Allard	TMM	Transfer Matrix Method
MST	Multiple Scattering Theory		

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Introduction

Context of the work

Noise is a major issue in our modern societies, ranging from everyday life inconveniences, work comfort problems to an actual pollution problem in severely affected areas. Sources of noise can be diverse; in urban environments, concerns are mostly on traffic and population noise. From industrial origins, noise can have various consequences, causing not only risks to the workers and the neighbouring populations, but also structural hazard to installations. Generally, motorized transport is a major issue in all environments, including aircraft, public transportation, roads, commercial vessel traffic, etc.

The European Environment Agency estimated in 2021 that 1 in 5 European resident are exposed to unhealthy levels of noise, its main cause being traffic. These noise exposures can have a strong impact on health¹ favoring impaired circadian rhythm, oxidative stress and cardiovascular dysfunction², as well as psychological consequences like anxiety and depression³. Moreover, anthropic noise pollution also affects biodiversity at large, causing stress, behaviour changes, and masking sensory perception⁴. Marine traffic was shown to generate stress on marine mammals and modify their behaviour by provoking area avoidance and vocalization disruption⁵.

Although sound intensity can be measured objectively, the impact of noise on individuals is in part shaped by social and cultural context. Studies show that expected behaviours norms and customs strongly influence reported annoyance⁶, with comparative surveys revealing clear cross-cultural differences. More recent work has linked collectivistic cultural norms to systematically higher everyday noise exposure as an example^{7,8}. Through these examples, we can conclude that noise is a very broad topic that raises complex questions in various scientific disciplines, regarding its impact on human life and biodiversity at large. In this work, we want to focus on noise mitigation using engineered structures in acoustics.

Most noise sources have emerged from the modernisation of societies through various innovations; with them come with the need to mitigate their noise impact up to our current capabilities, which is able to evolve thanks to research on noise barriers and acoustic insulation systems. Various strategies have been applied throughout the ages. Dating back to 4th century BC, acoustic jars were used to improve the room acoustics in Greek and Roman theatres as well as in the Middle Ages in Western Europe in churches⁹. These resonators are thought to be capable of reducing reverberation time in buildings, covering a frequency range matching that of human spoken voice. As another example, a famous roman amphitheater was shown to have a seat row disposition able to improve intelligibility of speeches from the stage¹⁰.

Porous materials

Porous and fibrous materials have been traditionally used to absorb sound in the last century^{11–16}, contemporarily to these antique devices. Largely used in the industry, they are efficient at frequencies starting close to the quarter wavelength resonance frequency of the layer and have been extensively analyzed in the literature^{17–21}. They can be found in natural occurrences or engineered and cover very large use cases of acoustic insulation such as industry, transport, aerospace, concert halls and many others.

These porous materials are lossy: an airborne acoustic wave propagating through a porous layer is dissipated as it propagates in the bulk, due to mainly two phenomena. Viscous friction occurs due to the drag between the air and the pore walls and another portion of the energy is converted to heat due to the small ducts and subsequently the existence of a thermal boundary layer. These losses mechanisms are strongly dependant on the nature of the material, as they are mainly affected by the geometry of the microstructure. Common models usually account for homogenized effective parameters to characterize these materials as will be discussed in a further section. Since wave velocity in porous media tends to be smaller than that of air, thus lowering slightly the frequency of the quarter-wavelength resonance and coincidentally corresponding to the frequency where such porous structure usually start to have an efficient acoustic absorption. Section 1.1 is dedicated to presenting the various models used for porous and poroelastic media modelling. Their absorption efficiency depends thus on their inherent losses but also on the impedance mismatch between the layer and the medium from which the impinging acoustic wave is originated (i.e usually air). The acoustic impedance mismatch leads to reflection at the interface, while the energy transmitted through the interface in the porous layer gets attenuated. In real-life applications, these materials are not closed systems and there is a need to account for their coupling to surrounding media into which part of the acoustic energy is susceptible to radiate.

A brief history of Acoustic Metamaterials

In more recent years, the emergence of more complex engineered structures has disrupted the landscape of acoustic absorbers. Metamaterials are composite structures designed to exhibit physical effective properties that go beyond the capability of the sole material that composes the bulk. They can be created by a lot of various means, including engineered microstructure, a gradial evolution of the material properties, or repeating patterns or scatterers implemented in one or several directions in space and embedded in a host material as shown for various dimensions in Fig. 1.1. Compared to homogeneous layers, they generally lead to more complex structures, and are able to achieve enhanced effects. Usually allowing for a tunable response, these structures offer great flexibility and open new perspectives in material design. This field has been initiated by Veselago, who has theoretically described the electrodynamics of materials having a negative refraction index (occurring in the case of a doubly-negative permittivity and permeability), now known as the Veselago lens in electromagnetic metamaterials²². These discoveries have subsequently sparked interest in various disciplines, such as photonics, optics²³ and in the case under consideration in this work, acoustics. Metamaterials can be used to design antennas and absorber for electromagnetic waves and seismic protection devices among others. In acoustic, they allow for various applications such as absorption, filtering, diffusion or

cloaking.

Seminal works have been done on the study of the use of lattice arrangements or crystals for wave control applications. Photonic crystals are used to achieve the control of light and electromagnetic waves²⁴. Similarly in acoustics, phononic crystals have been developed for the control of elastic and acoustic waves as the 2D (or 3D) periodic distribution of scatterers in a solid host medium²⁵ as depicted on Fig. I.1. In the case where the host medium is constituted by a fluid, these structures are called sonic crystals²⁶ and are often studied through their so-called band structure, also called dispersion diagram/relation when applied to the study of non-periodic geometries, which will be presented in Section ?? and extensively detailed Section 1.5. In the remaining of the literature overview, focus will be centered around metamaterials that involve periodicity in only one direction in most cases. These acoustic metamaterials can be divided into two main categories: the first concerns the so-called *reflection problem* where the case study is that of an incoming wave on some sample or panel which has a rigid-backing at the end. There, the consideration is focused on mitigating the reflected wave and achieving important acoustic absorption. The other case is that of a *transmission problem* where the incoming wave is able to be transmitted through the sample. In this case and depending on the application, either or both reflection and transmission are sought to be mitigated. In each of these cases respectively, the structures can be called metasurfaces and interfaces.

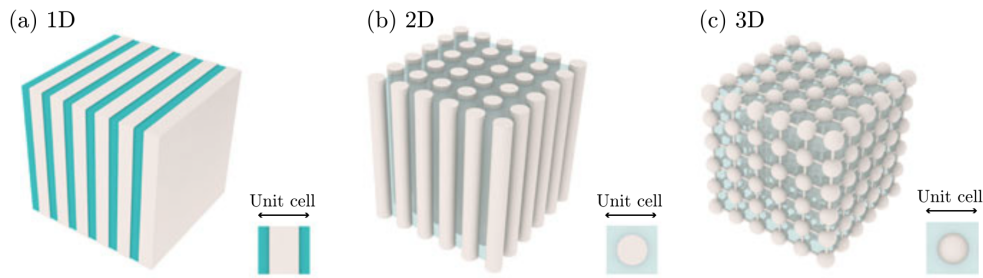


Fig. I.1: Periodic cells in 1, 2 and 3D. Image taken from Ref.²⁷

Structured materials that tackle the reflection problem can involve many various strategies. The combination of a perforated plate and coiled-up channels is able to achieve large broadband absorption underwater²⁸. Micro-perforated plate can also be used to achieve interesting absorption properties for airborne acoustics²⁹. Materials structured from resonant building blocks with resonance frequencies in the sub-wavelength regime i.e., for frequencies which wavelength is much bigger than the typical size of the material, have also been studied^{30–32}. In order to achieve these effects, such systems usually profit from the effects of bandgaps, which are frequency bands in the band structure where wave propagation is forbidden due to the destructive interference due to the presence of scatterers, have been achieved using locally resonant cells in the design of a sonic crystal³³. These resonances introduce strong dispersion of the waves and the properties of the system are thus dramatically affected. Together with the unavoidable losses, the resonant behaviour of metamaterials make them perfect candidates to design efficient absorbers at low frequencies with reduced dimensions^{34,35}. Balancing the dissipation and leakage of each resonance yields the recently described *critical coupling*. In this case, the reflection drops to zero³⁶. Perfect absorbers can thus be designed for reflection problems for both monochromatic waves^{34,37} and broad range of frequencies^{37–41}. This can

be achieved by tuning the structure using the scattering matrix zeros plotted in the complex frequency plane; critical coupling would be achieved when the scattering matrix zero has a vanishing imaginary frequency. Similarly to these broadband absorbers, Yang *et al.* have realized the backing of a porous layer using Fabry-Pérot channels to design an efficient structure⁴².

One of the initial interests of the acoustic metamaterials community was the possibility of producing deeps in the transmission spectra at very low frequencies. This has been widely developed in the literature, a small subset of these works include Refs. ^{43–46}. These configurations are based on resonant elements and display a zero transmission in the lossless case at their resonance frequency. In reality, losses cannot be avoided and a small amount of the energy is always transmitted. Membrane-type acoustic metamaterials^{47–49}, or acoustic metamaterials made by Helmholtz resonators^{50–52} have been used to develop systems with high values of transmission loss for audible acoustics. Considering the vibroacoustic coupling allows for more precise depiction of real-life applications. Works on periodic soft inclusion in a soft solid matrix was done by Aberkane *et al.*⁵³ showing improved transmission loss affected by the presence of soft inclusions in the solid layer. In this case, the presence of losses was able to improve the transmission loss at low frequencies. Also accounting for vibroacoustic coupling, attached resonators to walls have been also used to reduce the transmission loss of panels^{54,55}.

Mirror-symmetric structures are geometries that present a symmetry in the plane of the wave propagation meaning that a wave traveling from the right-side will be affected identically as if propagating from the left-side of the geometry. Their behaviour can be described using a two-port scattering process⁵⁶. It is worth noting here that if only a monopolar resonator (Helmholtz) or a dipolar resonator (membrane) is used to absorb waves, then only 50% of energy can be absorbed as shown in several contributions^{56,57}. However, using a material made of slits loaded by an array of identical resonators creates an accumulation point close to the resonance frequency of the resonators allowing the accumulation of the resonances of the slit, thus placing together symmetric and antisymmetric resonances^{47,50,58}, although requiring for a compromise between absorption and thickness. A more complex approach consists in designing degenerate resonators. By using combinations of membrane and cavity resonators^{47,59} and with slits loaded with Helmholtz resonators⁵⁷, quasi-total absorption can be reached. In these systems, each resonator manages half of the problem, absorbing each one half of the energy and allowing a perfect absorption peak: all the energy that impinges on the system is dissipated within the system, i.e., the energy is neither reflected nor transmitted. The other alternative to obtain perfect absorption consists in breaking the symmetry of the system. These systems can achieve unidirectional perfect absorption, i.e., perfect absorption can be obtained from one of the two incidence sides⁵⁶. The principle of these systems consists in using two resonators, one that is used to reduce as much as possible the transmission and the other that impedance matches the system from one side^{34,37,49}. This concept has been exploited to develop broadband perfect absorbers using a cascade of the previous process giving rise to a rainbow absorber³⁸.

From porous layers to metaporous surfaces

In order to increase the capabilities of acoustic absorption of porous materials, some works can involve the modification of the core structure of porous and poroelastic materials by tuning of the microstructure, introducing a gradient of the properties, or by using anisotropic materials. With the increase of new fabrication techniques based on 3D printing, the structure of the porous materials can be controlled and tuned to design materials with better absorption capabilities at low frequencies, based on gradient properties^{60,61}, or on anisotropic materials^{62–65}. Similarly, resonators filled with 3D printed porous materials have been developed to design perfect and broadband absorption⁶⁶. Other configurations have been studied, among them the periodic partitioning of the metaporous layer⁶⁷ or a labyrinthine channel embedded in a porous host⁶⁸, based on the idea of coiling up the space of the channel as was done for non-porous metamaterial designs. Recently, a hierarchical multilayer metaporous structure was proposed using an optimization procedure, giving rise to gradient and periodic designs⁶⁹.

Some studies involving porous layers embedding periodic arrays of inclusions were done during the last 15 years^{70–73}. These inclusions can have either resonances or not and they allow for an increased dissipation in the viscous regime. These structures are referred to as metaporous surfaces when used in reflection problems. The first works involving metaporous surfaces or interfaces were developed for transmission problems^{74,75}. In case of more than one inclusion per spatial period, several absorption peaks can be obtained for a particular arrangement along the porous thickness⁷¹. Nevertheless, this rapidly leads to a thick structure. Near the inclusion resonance frequency, chosen to be in the viscous regime, an absorption peak occurs⁷⁶. The resonance frequency of the inclusions depends on the orientation of the resonators relative to the rigid boundary^{72,76}. Recently, 3D printed porous materials with embedded split-ring resonators have been modeled⁶⁴, designed and experimentally characterized showing perfect absorption peaks. More complex spiral-shaped slits have been embedded in a porous layer and have shown to be able to maintain a high absorption for large incidence angles⁷⁷. Xiong *et al.* have studied a metaporous liner with embedded resonator and shown the existence of an exceptional point when the geometry is tuned properly, leading to enhanced absorption⁷⁸.

Previous metaporous layers studies have considered a host medium modelled as a rigid-frame porous layer, meaning that the layer is modeled as a lossy fluid if the elastic phase can be assumed motionless; extensions of these structures to consider poroelastic materials, that is considering the skeleton motion coupled to the fluid phase, have been done⁸². Interesting properties were exhibited for thin elastic shell inclusions, and particularly in the case of a coupling between the trapped and the first volume mode leading to a high wide band absorption. While this previous study implements a semi-analytical model that makes use of Multiple Scattering Theory (MST), it is worth noting here that different modelings exist as well, based on a Finite Elements Method (FEM) and MST have been validated for the case of an air-saturated metaporoelastic configuration with a solid inclusion and rubber coatings⁸³. This thesis is especially focused on the computation of the dispersion relation of such configurations. Therefore, a dedicated literature review dealing with the tools available to achieve such computation will be given in Chapter 1.

Wave propagation modelling and dispersion analysis

The analysis of wave propagation in periodic and complex media often relies heavily on numerical tools. Two main topics are of interest in this thesis: the scattering of an incident

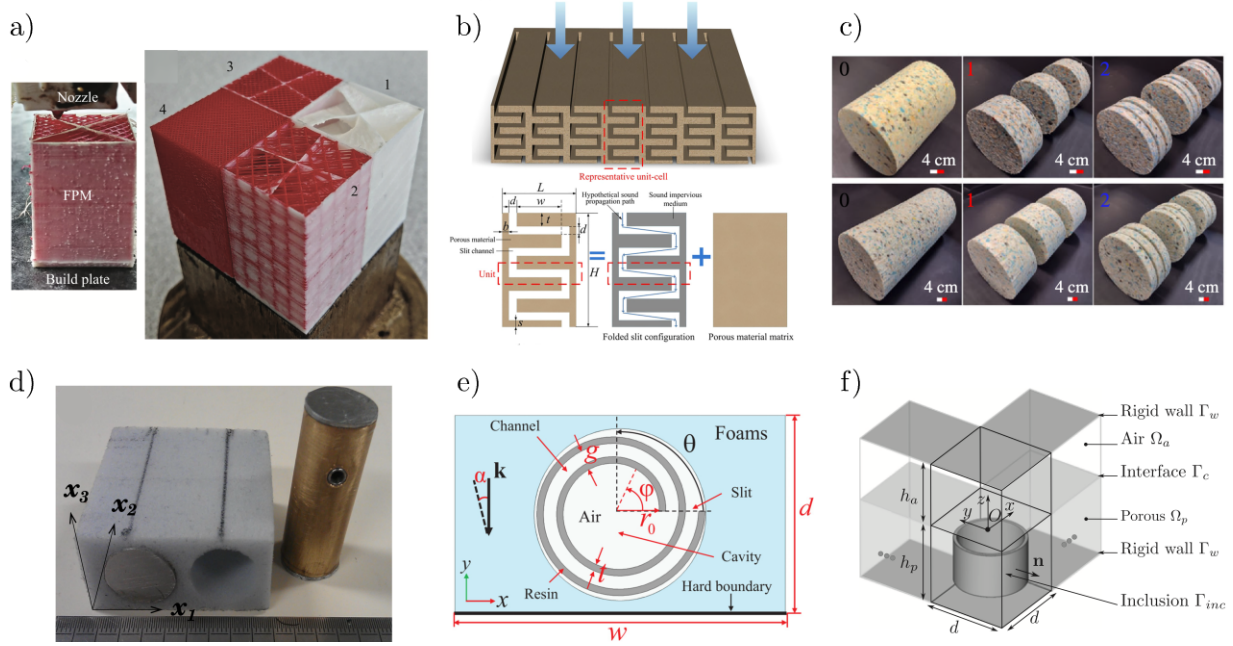


Fig. I.2: Example of metaporous structures: a) 3D printed microstructure with helocoidal shape⁷⁹, b) embedded labyrinthine channel⁸⁰, c) hierarchical configuration⁶⁹, d) Helmholtz resonators⁷⁶ e) archimedean spiral⁸¹ f) resonator tuned to an exceptional points⁷⁸

wave by a structure as well as its dispersion relation. The first one is simply the acoustic response of a system, where an incident wave propagates towards a specific geometry and is scattered by the latter. The scattering properties are of utmost importance, since tuning those remains one of the main topics for metamaterial applications^{27,31} and the goal is often to reach exceptional properties, such as perfect absorption of the incident wave by the structure. The scattering properties depend on the nature of the material(s) used, on the geometry and generally, the modal behaviour of the whole system.

In order to understand this behaviour in more details, we must turn to the study of the dispersion of the waves. Any causal system undergoing a scattering process can be described through their dispersion equation⁸⁴ as an algebraic expression. This allows for the characterisation of the dynamics of the modes in the system. Classically for dielectrics, the complex refractive index is expressed as a function of the frequency⁸⁵. In acoustics, we study dispersion in systems in the form $k(\omega)$, informing on the possible wavenumbers at which the system can respond for a given frequency. Reciprocally, the problem can also be written in the form $\omega(k)$, which is the other way around. From these equations, we can represent dispersion relations, diagrams that represents the wavenumber in a given direction against the frequency⁸⁶, thus informing on the evolution of the velocity of the propagation of waves and thus the dispersion and/or attenuation effects that the structure has. With these diagrams, the notions of phase velocity $v_\phi = \omega/k$ to qualify the local velocity of a wave and the group velocity $v_g = \partial\omega/\partial\text{Re}(k)$, defining the velocity of the envelope of a wavepacket or the velocity of the energy. In non-dispersive media, these quantities are the same, while the introduction of dispersion causes a distortion of a wavepacket with propagation⁸⁷.

In the context of metamaterials and periodic structures, dispersion relations are often referred to as band structure and have been widely used to characterize⁸⁸ sonic crystals⁸⁹. A

theoretical definition of a dispersion relation consists in the spectrum of all solutions of a set of equations given a set of parameters. shows the possible existing states of a system. Generally the form of the dispersion relation, and the various modes that are uncovered can give very in-depth details about the modal behaviour of any structure. As an example, in ultrasonics and non-destructive testing, localizing zero-group velocity (ZGV) points in the dispersion relation is very helpful for a number of uses including defects localization and material characterization⁹⁰.

Scarce literature focuses on the computation of the dispersion relations of waves propagating in dissipative layers with open domains general exists: the dispersion of waves in a single poroelastic layer has been studied by Boeckx *et al.*^{91,92} while an unpublished paper from Kiefer *et al.* took interest in Lamb waves in viscoelastic layer coupled to surrounding half-spaces⁹³. There is an interest in analogous studied of the coupling between surrounding media and metaporous surfaces, as part of the acoustic energy of the wave propagating in the surface should radiate back into the domain, leading to more complex loss effects on the structure.

These aspects will be discussed in detail in the following chapter, both from the theoretical perspective of porous and poroelastic media (and their coupling to surrounding media) and in terms of the available numerical tools for their computation.

Motivations and Contents of the thesis

This thesis is dedicated to the numerical modeling of wave propagation in open-system structures comprised of stacks of layers, homogeneous or with embedded scatterers. As it has been seen in the previous section, these materials combine the dissipative characteristics of porous media with the geometric or structural features typical of acoustic metamaterials, thus offering advanced control over their scattering properties. Understanding and predicting the acoustic behavior of such systems is therefore essential for designing efficient structures. In particular, this work focuses in part on the calculation of the dispersion of waves in these structures, and understand how the existing propagating and evanescent modes are shaping the scattering properties. Therefore, the development of efficient numerical strategies is critical to this study.

The tuning of metamaterials could be achieved in some studies by looking at the zeros of the reflection coefficient in the complex plane, thus highlighting the zeros of the scattering matrix. However, the wavenumber might be able to give out additional informations regarding the acoustic behaviour of metaporous layers. Band structure gives useful informations concerning sonic and phononic crystals. However, the dispersion relation of a metaporous layer embedded in an open system has never been addressed; both mechanisms of energy leakage in the surrounding media and dissipations in the porous layer should be investigated. However, this imply a heavy modification of the interface relations of the structure. It expected to have significant consequences on the propagation of modes therein. This work will be done on both homogeneous layers as well as periodic structures.

From these observations, we should review existing numerical methods able to compute the acoustic response of metaporous structures and consider ways to obtain a precise and robust method able to predict the scattering properties and acoustic responses of metaporous surface and interfaces. Additionally, we should evaluate methods that have been implemented for the computation of guided waves in layers and stacks of layers coupled to semi-infinite media.

Confronting dispersion relations against absorption coefficients should help to highlight the modal behaviour responsible for the various absorption peaks obtained by previously studied configurations and also highlight the full modes existing in the layers which might uncover additional phenomena.

This introductory section has set the context as well as an overview of the literature for the present thesis. We saw that acoustic metamaterials have emerged in the last decades and propose noise mitigation structures that are much more efficient than the traditional foams used for sound insulation. In particular, metaporous surfaces have shown to be promising options achieving one or multiple unitary absorption peaks at subwavelength frequency. Overall, a focus area has been identified towards identifying more precisely the modal behaviour of these configurations. Thus, this thesis focuses on two objectives: the development of numerical methods targeted to the specificities of metaporous surfaces and also on the computation of the dispersion relations of guided waves in these structures.

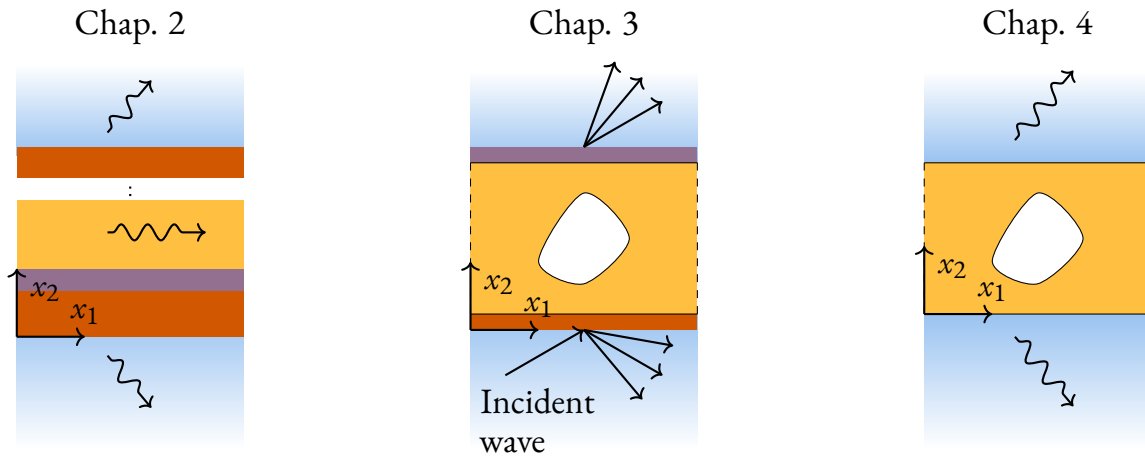


Fig. I.3: Diagram presenting a general layout for the configurations that will be studied in each chapter. The $\rightsquigarrow$ arrows represent radiative solutions while the $\longrightarrow$ arrows represent scattered amplitudes.

Organisation

The manuscript is split into four different chapters as follows,

- CHAPTER 1 introduces the fundamental concepts necessary for the good understanding of the thesis. This chapter focuses on the scattering of waves on dissipative layers and the guided waves formalism that is used for the calculation of dispersion relations. Additional bibliography is proposed on these topics as well as a summary of usable numerical methods to compute such problems. The chapter first introduces notions on the assumptions behind the widespread models of porous and poroelastic media. Next, wave propagation in systems made out of homogeneous layers is studied in the presence of an incident wave and the dispersion problem is studied, that is the free solution of the system of equations, without any forcing. The next sections deal with extension of the two scattering and dispersion solving for periodic unit cells that have embedded

scatterers. Finally, a summary of the overviewed methods is proposed with perspectives on the possibilities of our work.

- CHAPTER 2 targets the study of the propagation of guided waves in multilayer arbitrary structures with homogeneous dissipative layers. A Spectral Collocation Method (SCM) is implemented and extra steps allow for the local coupling of surrounding media allows for the computation of the dispersion relation for these configurations, along with the mode shapes and energy indicators. The formalism is presented in a general manner, so as to allow for any stack of elastic, fluid and poroelastic layers can be described with this method. Results are shown for a bilayer melamine - aluminium system accounting for the surrounding air, compared with a semi-analytical result and include the dispersion relation of the system, the energy flow at given location and the evaluation of the energy transport velocity for some of the branches. An experimental comparison is also proposed. This chapter is constituted from an article that has been published in Journal of Applied Physics^{P1}.
- CHAPTER 3 is focused on several numerical strategies for the modelling of scattering properties of metaporous/metaporoelastic structures. A plane wave approach is coupled to a FEM via a state vector formalism. Results focused on the performance of the proposed method, compared to other similar frameworks is proposed. The convergence of this method is shown for various metaporoelastic surfaces, which geometries were studied in previous papers. This chapter is a manuscript submitted to International Journal for Numerical Methods in Engineering^{P2}.
- In CHAPTER 4, the modal behaviour of metaporous structure is studied. The FEM method from the previous chapter is adapted to compute dispersion relations. Results are presented for the case of a rigid inclusion, as well as with embedded Split-Ring Resonator (SRR). Contents for this chapter are part of a draft that will be submitted for publication in Physical Review B.

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This chapter lays the foundation for the thesis by providing the fundamental theoretical notions and methods used in this work, in order to guide the reader through the various chapters that will follow. At first, an overview of the acoustic modelling of porous media is proposed. Following the presentation of the state of art in the introduction and of the areas of focus that have been identified from the latter, the key notions and methods for the computation of the scattering of waves by metaporous structures will be presented. From there, the formalism for the calculation of dispersion curves will be presented along with an overview of the methods used for such computations. Both of these topics will be illustrated with preliminary results in simple configurations.

1.1 Modelling of porous media

Porous media is a category of materials which encompasses naturally occurring substances such as rocks, soils and biological tissues, and engineered materials like ceramics, metallic foams, rockwool or certain types of polymers. They are considered biphasic, that is constituted of two distinct domains: a pore network Ω_f filled by a saturating fluid embedded in an elastic matrix Ω_s . A diagram of this microscopic structure is shown in Fig. 1.1. The porosity ϕ is defined as the fraction of fluid over the total volume of the material. In the limit case, $\phi = 0$ the material becomes purely a solid while at $\phi = 1$, it becomes a fluid. At microscopic scales, the geometry of the fluid phase thus forms a complex microstructure. Fortunately, at airborne acoustic scales, these materials can often be considered at a meso-/macroscopic level and therefore models have been developed to describe the large-scale motion of these materials. In this context, they can be regarded as homogeneous and described by a set of effective parameters that accounts for their complex geometry. The porous structure of these materials causes the incoming acoustic energy to be dissipated through thermal exchanges and friction (viscous losses) along the duct/pore network, while elastic dissipation take place when the skeleton and the fluid phase are coupled. These losses are thus dependant on the geometry, which parametrized by the porosity, the saturating fluid properties, the pore network size, connectivity and shape, as well as the frame mechanical parameters.

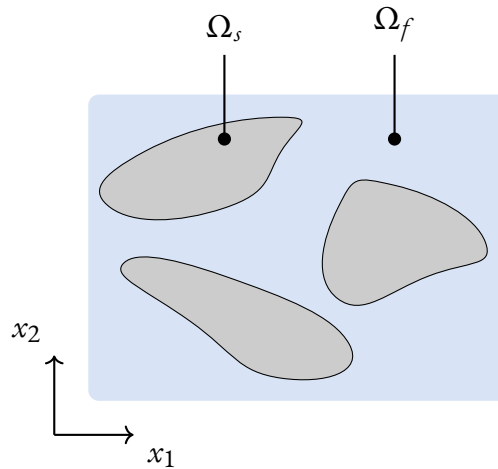


Fig. 1.1: Porous material structure

1.1.1. Rigid-frame approximation

In the so-called rigid-frame approximation, the skeleton of the porous material is considered motionless. Therefore, the pore walls are rigid walls and the material can be modeled as an *equivalent fluid* which possesses complex effective properties in order to model the losses mechanisms. A harmonic acoustic propagation is assumed with a positive time convention $e^{i\omega t}$ throughout this manuscript (with the exception of Chapter 2). Acoustic propagation is

governed by,

$$\nabla \cdot \left(\frac{1}{\rho_{eq}(\omega)} \nabla p(x, \omega) \right) = \frac{1}{K_{eq}(\omega)} \omega^2 p(x, \omega) \quad (1.1)$$

where p is the pressure field, ρ_{eq} is the effective density and K_{eq} is the effective bulk modulus. In our case (that of an isotropic and homogeneous medium), this equation can also simply be written as an Helmholtz equation as $(\nabla^2 + k_{eq}^2) p(x, \omega) = 0$ with $k_{eq} = \omega \sqrt{\rho_{eq}/K_{eq}}$ the equivalent fluid wavenumber.

The effective medium parameters are usually calculated using the Johnson-Champoux-Allard (JCA) semi-phenomenological model^{94,95}, developed from 1987 to 1991, that accounts for the viscous and inertial dissipative effects. The materials parameters are thus complex and frequency-dependant; the density accounts thus for the viscous effects while the bulk modulus accounts for thermal exchanges. These two quantities are given by the JCA model as,

$$\begin{aligned} \rho_{eq}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi} \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \omega} \sqrt{1 - i \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda} \right)^2} \right), \\ K_{eq}(\omega) &= \frac{\kappa P_0}{\phi} \left(\kappa - (\kappa - 1) / \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \text{Pr} \omega} \sqrt{1 - i \text{Pr} \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right) \right)^{-1}. \end{aligned} \quad (1.2)$$

These expressions involve a large number of effective parameters for the porous material; α_∞ is the tortuosity, R_f is the flow resistivity, Λ and Λ' are respectively the viscous and thermal characteristic length. The saturating fluid is characterized in this model by its heat capacity ratio κ (commonly also denoted γ), density ρ_f , kinematic viscosity μ_f , pressure P_0 and Prandtl number $\text{Pr} = c_p \mu_f / k$ with c_p is the specific heat and k the thermal conductivity. We can introduce also the Biot frequency $\omega_c = R_f \phi / (\rho_f \alpha_\infty)$ that defines the transition frequency between the inertial and viscous regimes.

1.1.2. Elastic motion of the skeleton: Biot theory

While the equivalent fluid formalism is able to accurately model porous material in specific sets of conditions, that is when the skeleton can be assumed to be motionless, this assumption can quickly become limited as soon as the said skeleton is subjected to an elastic deformation. This is can be the case when the saturating fluid of the PEM has a high density (e.g. water), when the skeleton is coupled to elastic plates or when the acoustic radiation has a wavelength comparable to the pore scale, although these are just examples. In this situation, the full elastic motion should be modeled, so that to account for the coupling between the solid matrix and its fluid filling. The Biot theory^{96–99} is a formalism that encompasses the theory of the deformation of porous material saturated by an incompressible fluid, used both in geophysics and acoustics. For sake of conciseness, the various models will be briefly be presented along with the relevant physical quantities. A full derivation would be out of scope and is presented in existing works¹¹.

Before presenting the full Biot model, some quantities should be provided additionally to the equivalent fluid parameters. The frame is assumed to undergo elastic deformations, thus

the common mechanical parameters used are the solid density of the frame material ρ_s as well as its apparent density $\rho_1 = (1 - \phi)\rho_s$. The bulk moduli are denoted K_b and K_s , respectively for the frame and the material. The shear modulus is also considered complex, accounting for structural losses as $\tilde{N} = N(1 + i\eta)$ with η the loss factor. Symbols written with a tilde denote complex expressions and therefore account for some sort of loss mechanisms. For the fluid, the apparent density is $\rho_2 = \phi\rho_f$ and we define $\rho_{12} = -\phi\rho_f(\alpha_\infty - 1)$. A summary on Biot theory is also proposed in Chap. 2) with the tilde symbols omitted for ease of reading.

Formulations for the motion equations

The Biot poroelastic equations from the original 1956 formulation^{96,98,99} depend on the solid displacement vector $\mathbf{u}_s$ and the fluid displacement $\mathbf{u}_f$,

$$\begin{aligned}\nabla \cdot \boldsymbol{\sigma}_s(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{11} \mathbf{u}_s - \omega^2 \tilde{\rho}_{12} \mathbf{u}_f, \\ \nabla \cdot \boldsymbol{\sigma}_f(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{12} \mathbf{u}_s - \omega^2 \tilde{\rho}_{22} \mathbf{u}_f,\end{aligned}\quad (1.3)$$

where $\boldsymbol{\sigma}^s$ is the partial solid stress tensor, $\boldsymbol{\sigma}^f$ is the partial fluid stress tensor and the apparent densities of the solid-phase is $\tilde{\rho}_{11}$, that of the fluid is $\tilde{\rho}_{22}$ and the coupling apparent density is $\tilde{\rho}_{12}$. They are complex quantities expression that also account for the contribution of viscous forces and are given as,

$$\tilde{\rho}_{22} = \phi^2 \tilde{\rho}_{eq}, \quad \tilde{\rho}_{12} = \rho_2 - \tilde{\rho}_{22}, \quad \tilde{\rho}_{11} = \rho_1 - \tilde{\rho}_{12}. \quad (1.4)$$

These coupled motion equations are associated to the stress-strain relation,

$$\begin{aligned}\sigma_{s,ij} &= [(\tilde{P} - 2\tilde{N})\nabla \cdot \mathbf{u}_s + \tilde{Q}\nabla \cdot \mathbf{u}_f] \delta_{ij} + 2\tilde{N}\epsilon_{s,ij}, \\ \sigma_{f,ij} &= (\tilde{Q}\nabla \cdot \mathbf{u}_s + \tilde{R}\nabla \cdot \mathbf{u}_f) \delta_{ij} = -\phi p \delta_{ij},\end{aligned}\quad (1.5)$$

where $\nabla \cdot \mathbf{u}_s$ and $\nabla \cdot \mathbf{u}^f$ correspond to the dilatation of the solid and fluid phases respectively and ϵ_s is the solid-phase strain tensor. The Biot coefficients are denoted as $\tilde{P}$ for the solid elastic coefficient, $\tilde{Q}$ for the elastic-fluid coupling coefficient, and $\tilde{R}$ for the fluid-phase coefficient. In the context of a PEM saturated by a light fluid, their expressions are

$$\tilde{P} = K_b + \frac{4}{3}N + (1 - \phi)^2 \frac{\tilde{K}_{eq}}{\phi}, \quad \tilde{Q} = \tilde{K}_{eq}(1 - \phi), \quad \tilde{R} = \phi \tilde{K}_{eq}. \quad (1.6)$$

Generalized forms of these coefficients are required when the limit $K_b/K_s \rightarrow 0$ no longer holds, as is the case for water-saturated PEM, and are given in Ref.¹¹.

A different formulation, especially convenient for finite-elements discretization is known as the $(\mathbf{u}^s, p)$ formulation. First, new parameters in terms of the Biot elastic coefficients and apparent densities can be defined,

$$\hat{A} = \tilde{P} - 2\tilde{N}_s - \frac{\tilde{Q}^2}{\tilde{R}}, \quad \tilde{\rho} = \tilde{\rho}_{11} - \frac{\tilde{\rho}_{12}^2}{\tilde{\rho}_{22}}, \quad \tilde{\gamma} = \phi \left(\frac{\tilde{\rho}_{12}}{\tilde{\rho}_{22}} - \frac{\tilde{Q}}{\tilde{R}} \right). \quad (1.7)$$

γ is another coupling coefficient, also denoted as $\tilde{\beta}$ in the derivations of Chapter 2. Using the second row of Eq. (1.3) allows for the expression of the fluid displacement as, $\mathbf{u}^f =$

$\phi/(\tilde{\rho}_{22}\omega^2)\nabla p - \tilde{\rho}_{12}/\tilde{\rho}_{22}\mathbf{u}^s$ and using that in Eq. (??) and Eq. (1.5) allow to rewrite the coupled motion equations. Using these, the so-called $(\mathbf{u}_s, p)$ formulation of the Biot theory¹⁰⁰ is expressed as,

$$\begin{aligned}\nabla \cdot \hat{\boldsymbol{\sigma}}_s + \tilde{\rho}\omega^2\mathbf{u}_s + \gamma\nabla p &= 0, \\ \frac{\nabla^2 p}{\rho_{22}\omega^2} - \frac{\gamma}{\phi^2}\nabla \cdot \mathbf{u}_s + \frac{1}{R}p &= 0,\end{aligned}\tag{1.8}$$

where $\hat{\boldsymbol{\sigma}}_s$ is the *in vacuo* solid stress tensor,

$$\hat{\sigma}_{s,ij} = \sigma_{s,ij} - \phi \frac{\tilde{Q}}{R} p \delta_{ij} = \hat{A} \nabla \cdot \mathbf{u}_s \delta_{ij} + 2\tilde{N} \epsilon_s.\tag{1.9}$$

Propagation in unbounded poroelastic media

Poroelastic media modelling supports three waves: a compression φ_p and shear wave ψ_s related to the solid phase as well as an additional compression wave φ_a , related to the fluid phase. In the case of each of these formulations, the fields can be expressed depending on these potentials,

$$\mathbf{u}_s = \nabla \varphi_p + \nabla \varphi_a + \nabla \times \psi_s, \quad \mathbf{u}_f = \mu_p \nabla \varphi_p + \mu_a \nabla \varphi_a + \mu_s \nabla \times \psi_s,\tag{1.10}$$

with μ_p, μ_a, μ_s are coefficients determined by the ratio of solid over fluid displacements ratio. The wavenumbers for each of these waves are, following section 6.5 of Allard and Atalla textbook¹¹ and denoted as $\delta_i, i = 1, 2, 3$ in the text,

$$\begin{aligned}k_{p,a} &= \sqrt{\frac{1}{2} \left(\delta_2^2 + \delta_{eq}^2 \pm \delta_t \right)}, \\ k_s &= \omega \sqrt{\frac{\hat{\rho}}{N}},\end{aligned}\tag{1.11}$$

respectively for the solid compression, fluid compression and shear wave, with,

$$\delta_{eq} = \omega \sqrt{\frac{\tilde{\rho}_{eq}}{K_{eq}}}, \quad \delta_1 = \omega \sqrt{\frac{\hat{\rho}}{P}}, \quad \delta_2 = \omega \sqrt{\frac{\hat{\rho}_s}{P}}, \quad \delta_t = \sqrt{(\delta_2^2 + \delta_{eq}^2)^2 - 4\delta_{eq}^2 \delta_1^2}.\tag{1.12}$$

The propagation characteristics, i.e. wavenumbers of the waves, of porous and poroelastic media are illustrated in Figure 1.2. As discussed in the previous sections, the wavenumber for the equivalent fluid as well as the three for the poroelastic layer are complex-valued. Their real part account for the propagation of the phase flow, inversly proportional to the wave velocity and their imaginary part accounts for the attenuation that occurs due to the losses inherent to each of these materials. It appears that at low frequencies, the attenuation is very low for all waves. The velocity of the compression and shear waves (observed on the figures by the slope of the real parts of the wavenumbers) is much higher than the waves in the fluid for the equivalent fluid and the compression fluid phase velocity, as they propagate in a soft solid where the velocity is usually much higher than in a fluid.

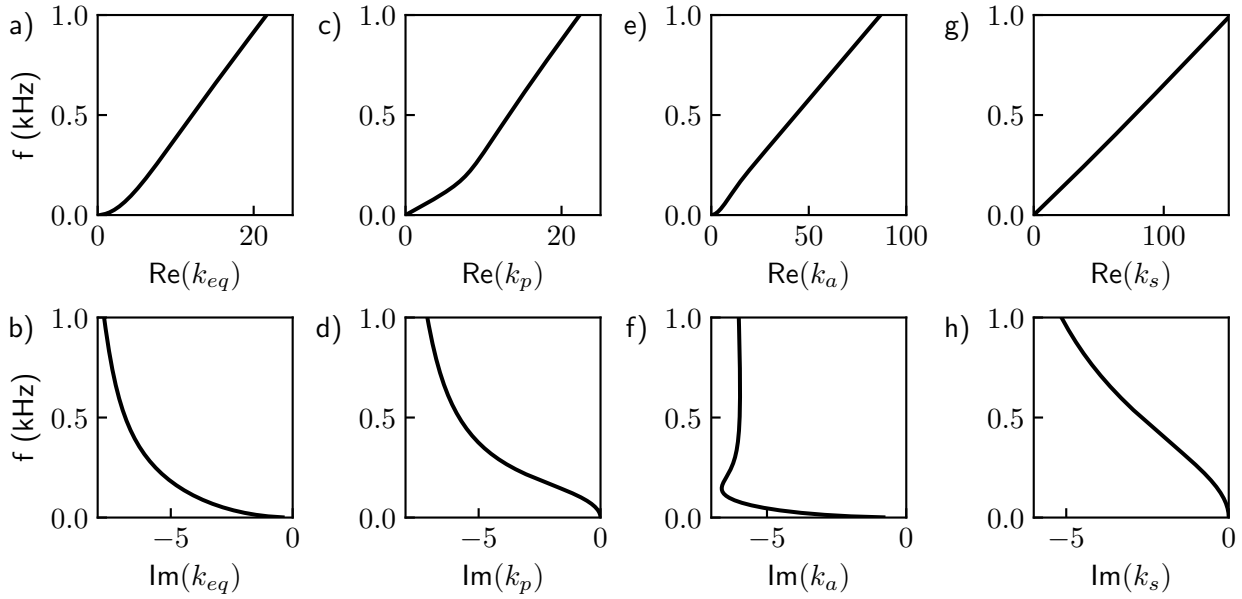


Fig. 1.2: Wavenumbers of waves in unbounded porous media in the case of an a-b) equivalent fluid and c-h) for the Biot model.

1.2 Wave propagation in homogeneous layers

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of the wave propagation in unbounded porous and poroelastic media have been shown. The present section proposes the introduction of a geometry, that is to say boundaries or interface conditions delimiting different domains characterized each by a different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, impinging at the surface of a structure and is then scattered (reflected and transmitted) from and through each interface. This is illustrated in Fig. 1.3 for the two main cases that are the so-called reflection problem where the structure is bounded on one side by a rigid-backing and the transmission problem where the incident wave is reflected and transmitted through the layer.

If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is referred to as a layer. The waves supported by the medium propagate by reflecting alternatively from one surface to the other; they are said to be guided along the layer. These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often lead to writing complex systems that can be difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries. In a first section, the scattering of a plane wave by a homogeneous porous layer is investigated along with the various methods available. Next, the methods for the computation of the modes of these configurations are presented. In a subsequent section, more complex geometries, those of metaporous structures, are of interest and both methods for the computation of the scattering coefficients and the modes of those configurations will be discussed.

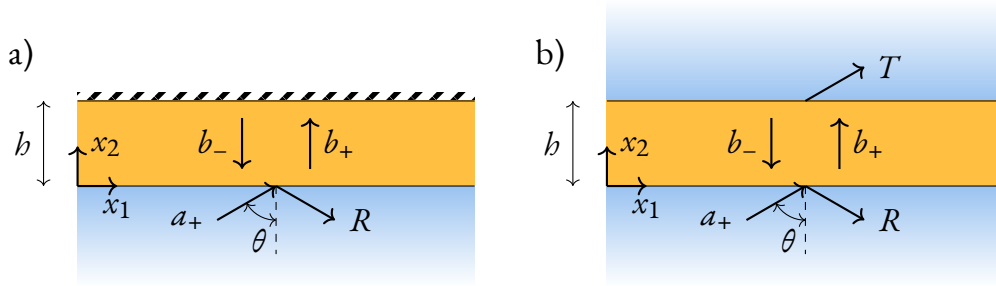


Fig. 1.3: Diagram representing the cases of the a) reflection and b) transmission problem

1.2.1. Analytical plane wave modelling

Modeling acoustic propagation in homogeneous layers by assuming plane waves is the simplest formalism in acoustics. We assume these plane waves such that they satisfy the motion equations, e.g. the ones introduced in eq. (1.1) or eq. (1.3). Introducing a simple problem, that of the scattering of an incident plane by a porous layer, we present an example of plane wave modelling for a simple system. The system is 2D with coordinates $\mathbf{x} = (x_1, x_2)$ constituted of a porous layer with thickness h and thus two interfaces of infinite extent along x_1 , there is a fluid-porous interface at $x_1 = h$ and a porous-fluid interface at $x_1 = 0$. Both media of the air half-spaces and porous layer are modeled as fluid, thus the interfaces conditions should impose continuity of the pressure fields and normal particle displacements. The incident pressure field that satisfy the Helmholtz equation and originated from the lower interface takes the form of,

$$p_{0-} = a_+ e^{-ik_1 x_1} e^{-ik_{2,0} x_2} + R e^{-ik_1 x_1} e^{ik_{2,0} x_2}, \quad (1.13)$$

with a_+ the incident wave amplitude and R , the amplitude of the reflected wave. The pressure field in the porous layer is expressed as a sum of forward and backward waves,

$$p_p = b_+ e^{-ik_1 x_1} e^{-ik_{2,eq} x_2} + b_- e^{-ik_1 x_1} e^{ik_2(x_2-h)}, \quad (1.14)$$

and at the upper interface,

$$p_{0-} = T e^{-ik_1 x_1} e^{-ik_{2,0}(x_2-h)}, \quad (1.15)$$

with T the transmitted wave amplitude. All wavenumbers should hold for the Snell-Descartes law that states

$$k_1^2 + k_{2,0}^2 = k_0^2, \text{ and } k_1^2 + k_{2,eq}^2 = k_{eq}^2. \quad (1.16)$$

The wavenumber should be imposed with $k_1 = k_0 \sin \theta$, $k_{2,0} = -k_0 \cos \theta$, with θ the angle of incidence, that governs the direction of propagation of the incident field. The displacement fields are additionally expressed using the momentum equation relating the latter with the pressure fields as $\partial_2 p = -i\omega \rho_i u_2$, with $i = 0, eq$.

Next, the interface conditions describe the relationships between properties at the boundary between two media, including the continuity or vanishing of given fields. If a media is terminated by a surface (e.g. a rigid wall), a constraint should be applied on the fields to represent it, denoted as boundary conditions. Assuming a continuity between the two fluids in the present case with interfaces located at $x_2 = 0$ and $x_2 = h$, we obtain the following conditions,

$$p^t(h) = p^l(h), \quad u_2^t(h) = u_2^l(h) \quad p^l(0) = p^b(0) \quad u_2^l(0) = u_2^b(0), \quad (1.17)$$

with superscripts b and t denoting respectively the bottom and top interfaces.

The full analytical description of a system constituted from one or several interfaces can be done by writing a linear system built from the interface conditions, which vector of unknowns are the amplitudes of each of the existing waves in the system, that is $(b_+ \ b_- \ R \ T)^T$. The forcing right-hand side corresponds to the incident wave terms. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients. Imposing the interface conditions leads to writing the following matrix system,

$$\begin{pmatrix} 1 & e^{ik_{2,eq}b} & 0 & -1 \\ \frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,eq}}{\rho_{eq}}e^{ik_{2,eq}b} & 0 & \frac{k_{2,0}}{\rho_0} \\ e^{ik_{2,eq}b} & 1 & -1 & 0 \\ \frac{k_{2,eq}}{\rho_{eq}}e^{ik_{2,eq}b} & -\frac{k_{2,eq}}{\rho_{eq}} & -\frac{k_{2,0}}{\rho_0} & 0 \end{pmatrix} \begin{pmatrix} b_+ \\ b_- \\ R \\ T \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ a_+ \\ \frac{k_{2,eq}}{\rho_{eq}}a_+ \end{pmatrix}. \quad (1.18)$$

From this expression, the system can simply be solved to obtain all scattered amplitudes b_+ , b_- , R and T , where R and T correspond respectively to the reflection and transmission coefficient. In the case of a rigidly-backed layer, the interface at $x_2 = b$ is replaced by a rigid termination. For (equivalent) fluids, a hard wall or Neumann-type boundary condition is translated by a vanishing normal particle displacement field along the boundary, that is $u_2(\Gamma) = 0$. The term depending on T thus vanishes, since p_{0+} does not exist; this reduces the size of the system.

This method is denoted in this work as the Global Method (GM), since all of the available data is used in the formulation of the problem as the unknowns are all wave amplitudes. As described in Section ?? of Chapter ??, a general formalism can be used for this method and any stack of any layer can be described with the GM, by putting together a matrix system that contains all interface conditions in the system, in the general form,

$$\mathbf{M}(\omega)\mathbf{x} = \mathbf{f}, \quad (1.19)$$

where $\mathbf{M}$ corresponds to the interface conditions involved in the system, $\mathbf{x}$ contain the wave amplitudes and $\mathbf{f}$ is the forcing term, the incident plane wave amplitudes.

Similarly, a linear system can be written in the case of a PEM layer; the outer pressure fields remain the same but a solid and a fluid displacement field is modeled in the layer, added to the pressure field, as mentioned in Section 1.1. The interface conditions for a poroelastic-fluid coupling are thus,

$$\sigma_{12}^s = 0, \quad \sigma_{22}^s + \sigma_{22}^f = -p_0, \quad \sigma_{22}^f = -\phi p_0, \quad \phi u_2^f + (1 - \phi)u_2^s = 0. \quad (1.20)$$

and for a rigid-backing,

$$u_1^s(\Gamma) = 0, \quad u_2^s(\Gamma) = 0, \quad u_2^f(\Gamma) = 0. \quad (1.21)$$

For the case of a rigidly-backed layer, the system is entirely written in in the appendices of Ref.⁸². For the symmetric layer, the linear system is given in Ref.⁹². Finally, the procedure to obtain the scattering coefficients is the same as previously.

1.2. Wave propagation in homogeneous layers

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in the case of a symmetric configuration (with air coupling at both interfaces) are represented by the diagrams in Figs 1.4a and b and results in Figs 1.4c and d, respectively. The GM results are presented by the continuous lines. There, the layer for both porous and PEM are both $h = 5$ cm thick and the incident wave propagates with a $\theta = 30^\circ$ angle. The parameters for each material corresponds to the one used in Table 2.1 from Chapter ??.

The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but additional effects due to the skeleton motion can be observed on the continuous lines while the equivalent fluid response shows to be smoother. For all configurations, the melamine layer is acoustically transparent at lower frequency and absorption can be noticed at slightly higher frequencies.

1.2.2. Transfer Matrix Method

The widely used method we know today as the Transfer Matrix Method (TMM), was initially developed by the works of Thomson¹⁰¹ and Haskell¹⁰² in order to describe elastic wave propagation in multilayer systems. This method can be applied to a range of wave physics among geophysics, quantum mechanics, electromagnetics and obviously acoustics^{markos2008, 12,27} and elastic waves for ultrasonics¹⁰³. Its principle lies in the evaluation of the state vector that regroups the variables of interest in the system and evaluate them at each interface, which is possible thanks to the interface conditions. Then, the physical field amplitudes stored in the state vector are propagated from one layer to the next in the case where multiple coupled layers are modeled. The transfer matrix T of a fluid layer, relating the pressure and normal displacement fields from the bottom (located at $x_2 = 0$) to the top interface (located at $x_2 = h$) of a fluid layer is thus,¹²

$$\underbrace{\begin{pmatrix} p(0) \\ u_2(0) \end{pmatrix}}_T = \begin{pmatrix} \cos(k_{eq}h) & iZ_{eq} \sin(k_{eq}h) \\ iZ_{eq}^{-1} \sin(k_{eq}h) & \cos(k_{eq}h) \end{pmatrix} \begin{pmatrix} p(h) \\ u_2(h) \end{pmatrix}, \quad (1.22)$$

where k_{eq} is the bulk wavenumber of the wave propagating in the equivalent fluid and Z_{eq} is the medium impedances. The extension to PEM makes of a state vector that includes the fields, $(\hat{\sigma}_{12} \ u_2^s \ u_2^t \ \hat{\sigma}_{22} \ p \ u_1^s)^T$. More details are given in Chapter 11 of Allard and Atalla textbook¹¹.

Stacks of N layers that have the same material nature are easy to manipulate, the transfer matrix of each layer has the same size as the others and thus,

$$T_{\text{global}} = \prod_{n=0}^N T_n. \quad (1.23)$$

TMM allows for an easy multiphysics coupling with interface matrices that relate the relations between the various fields at an interface which are thus non square matrices. This method was deep dived in several textbooks among which we propose Refs.^{11,27} for additional details. In the last decades, there has been an effort in extending this method to stabilize it, and various formulations have been introduce to achieve various stuff [citer des trucs jsp, global matrix method \(knopoff1964](#) (L. Knopoff, "A matrix method for elastic wave problems," *Bull. Seism.*

Soc. Am., vol. 54, pp. 431–438, 1964.)), **admittance matrix method**. The TMM also allows for the computation of complex system by an assembly of elements including admittance matrices in order to consider a Helmholtz Resonator (HR) embedded in an equivalent fluid layer¹⁰⁴.

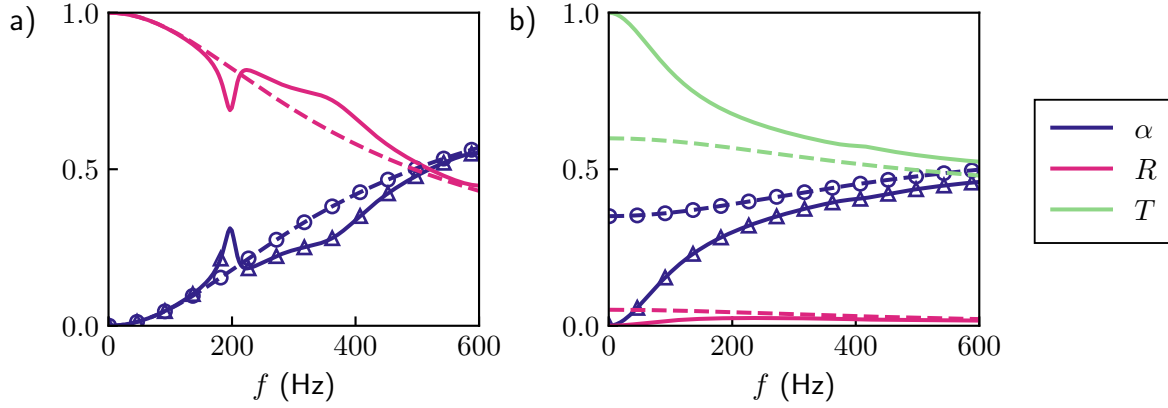


Fig. 1.4: Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) computed using the GM for the a) reflection, b) transmission problem. Results computed with the TMM are also shown for the absorption coefficient for the equivalent fluid (o) and for the poroelastic layer (Δ)

While very versatile, it remains unstable in limit cases at high frequencies or for large thickness of the layers. While the formulation is mathematically exact, the evaluation of the system involves finite-precision calculation which leads the very sensitive exponential terms to diverge in the previously specified regimes. From the evaluation of the transfer matrix of the global system, the scattering matrix can be obtained and thus the reflection and transmission coefficients by simply manipulating the expressions for the fields contained in the state vector with the scattering coefficients.

Considering a rigidly-backed layer, and thus $u_2(h) = 0$, we obtain a reflection coefficient,

$$R_b = \frac{T_{11} - Z_{eq}T_{21}}{T_{11} + Z_{eq}T_{21}}. \quad (1.24)$$

In the case of the symmetric layer, the reflection and transmission coefficients are determined from the scattering matrix as well,

$$T_s = \frac{e^{-ikL}}{T_{11} + T_{12}/Z_{eq} + Z_{eq}T_{21} + T_{22}}, \quad R_s = \frac{T_{12}/Z_{eq} - Z_{eq}T_{21}}{T_{11} + T_{12}/Z_{eq} + Z_{eq}T_{21} + T_{22}}. \quad (1.25)$$

which is only valid for reciprocal and linear assumptions. The absorption coefficient for the same two configurations are calculated as $\alpha_b = 1 - |R_b|^2$ and $\alpha_s = 1 - |R_s|^2 - |T_s|^2$ and are shown in Fig. 1.4 with markers. It can be seen that the same results (up to the respective numerical precision of each method) are obtained. No absorption peak is observed in the results since the frequency range is way below the quarter-wavelength resonance (which should lie around 1.5 kHz for these 5 cm-thick homogeneous layers). The equivalent fluid and the full Biot theory results follow generally the same trend for the results of the rigidly-backed layer in Fig. 1.4a, except for a peak at 200 Hz for the poroelastic model. This is known as the Biot peak and involves elastic effects.^{zielinski2011, 82}. The results in the transmission case in Fig. 1.4b present a

generally lower absorption, bounded at $\alpha = 0.5$. Here, the equivalent fluid and poroelastic respective response are significantly different: at low frequencies, the Biot theory behaves as expected with the transmission close to 1 and the reflection close to 0. The equivalent fluid reflection coefficient is similar but its transmission lies around 0.6 for lower frequencies, leading to a rather important absorption coefficient at zero frequency. While not representing a physical and observable behaviour of the system, this effect is a feature of the JCA model. [demander infos à OD, JPG](#).

1.3 | Guided waves for dispersion problems

In this section, we take an interest in similar configurations as previously, removing the excitation or forcing term in the equations. Solving the appropriate set of equations for a given set of parameters leads to the modes of the free system, that is the solution of the characteristic equation leading a specific behaviour of the wave propagation in the structure for the said parameters. The form of the modes of wave propagation in layer is discussed and various methods for their calculation of are presented.

The existence of an interface in the geometry leads to the propagation of not only the bulk waves but also of so-called surface waves¹⁰⁵ where the interface itself acts as the support for propagation. A dispersion relation can be illustrated using the example of a fluid layer of infinite extent in one direction and Neumann-type boundary conditions at each end $x_2 = 0$ and $x_2 = h$. This leads to modes of the system for solutions of $\sin(kh) = 0$,¹² from which we obtain $k_n = n\pi/h, \forall n \in \mathbb{N}$. Finally the dispersion relation writes as,

$$k_n(\omega) = \sqrt{\left(\frac{\omega}{c_0}\right)^2 - \left(\frac{n\pi}{h}\right)^2}, \quad (1.26)$$

with c_0 the velocity in air. The solutions to this equation for a range of $n = 0$ to $n = 4$ is shown in Fig. 1.5. There, the wavenumber appears as either purely real or purely imaginary, indicating respectively the propagative and evanescent modes of this simple geometry. The mode with order 0 is the one that exists in unbounded fluid materials, with its slope corresponding to the fluid velocity c_0 . The next orders all present a cut-off frequency below which the modes become evanescent.

Different types of waves propagate along the interface. An example can be water waves that follow the interface between air and water; most of the energy is localized at the interface. A similar behaviour exists for seismic waves where the interface of a solid substrate supports the propagation of Rayleigh (longitudinal motion) and Love waves (transverse motion) which both show an energy strongly decaying with depth. Scholte waves are waves that propagate along the interface between a fluid and a solid medium. In fluid layers, the propagating waves are similar to studying a shear horizontal (SH) wave in an elastic layer. One mode propagates at zero frequency, while countless other modes propagate with a cut-off frequency that corresponds to the successive quarter-wavelength resonances, essentially depending on the layer thickness and the nature of the boundary/interface conditions.

From the previously obtained linear systems in Section 1.2.1 for the configurations, and removing the incident wave from the system, i.e. the forcing (right-hand side) term, an homogeneous linear system is obtained, in the general form $\mathcal{L}(\mathbf{k}, \omega)\mathbf{s} = 0$, where the operator $\mathcal{L}$

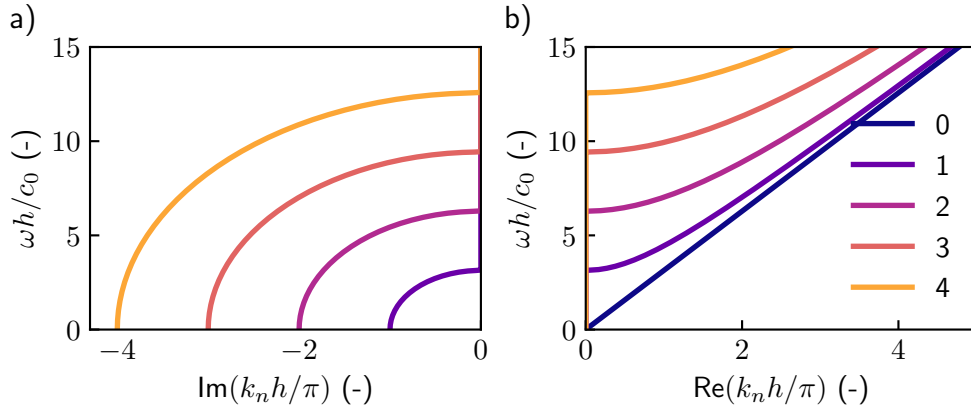


Fig. 1.5: Adimensional dispersion relation of a layer of air with $b = 5$ cm with a) imaginary parts of the wavenumber and b) real parts of the wavenumber

governs wave propagation in the structure, $\mathbf{k} = (k_1, k_2)^T$ is the wavenumber vector in the 2D geometry and $\mathbf{s}$ a vector containing the amplitude of all the partial waves. In order to obtain nontrivial solutions for $\mathbf{s}$, the system should satisfy a vanishing determinant for the linear system, $\det(\mathbf{L}(\mathbf{k}, \omega)) = 0$. The parameters satisfying this so-called *dispersion equation* can be sought, allowing to identify the modes of the structure. These equations can be solved for various parameter spaces depending on the perspective adopted on the problem^{106,107}: solving for frequencies while imposing the wavenumber or the other way around. The first one is useful when interested in the time-dependant effects the structure has on the propagation. In this work, we are interested in the losses generated by the structure and the material and as such frequency is fixed in order to solve complex wavenumbers. The solutions are thus able to indicate on rather the spatially induced dispersion occurring in the system.

1.3.1. A bestiary of wave solutions in dissipative and open waveguides

In this section, an overview of the possible sets of solutions for the guided waves in waveguide is presented. These sets of solutions are determined by the direction of the propagation, the attenuation and the nature of the leaky wave if it exists. This *bestiary* is split into 3 categories. Considering a layer and half-space system, there exists three wavenumber components from Eq. (1.16). The form of solutions are assumed as

$$p_l \propto e^{-ik_1 x_1} e^{\pm i k_{2,eq}}, \quad p_0 \propto e^{-ik_1 x_1} e^{\pm i k_{2,0}}, \quad (1.27)$$

with $k_1 = +|k'_1| - i|k''_1|$ and $k_{20} = \pm|k'_{20}| \pm i|k''_{20}|$. The longitudinal wavenumber is the same in both layer and half-space domain as a consequence of the Snell-Descartes law and the continuity equations: there exists in total three dependant wavenumber components in this system, k_1 , $k_{2,0}$ and $k_{2,eq}$ among which determining two allows to compute all,

$$k_{2,0} = \pm \sqrt{k_0^2 - k_1^2}, \quad k_{2,eq} = \pm \sqrt{k_{eq}^2 - k_1^2}. \quad (1.28)$$

An important relation between these wavenumbers can be obtained by taking the imaginary part of Eq. 1.16,

$$k'_1 k''_1 = -k'_{20} k''_{20}. \quad (1.29)$$

1.3. Guided waves for dispersion problems

A set of physically meaningful types of solutions should satisfy this equation,¹⁰⁸; the following section will provide a detailed discussion of this issue.

Because of the symmetrical nature of the presented configurations, and the considerations of unbounded layers in the x_1 direction, a plane of solutions corresponding to $k'_1 \geq 0$ can be selected, as the right-going solution is perfectly equivalent solutions to the left-going solutions.

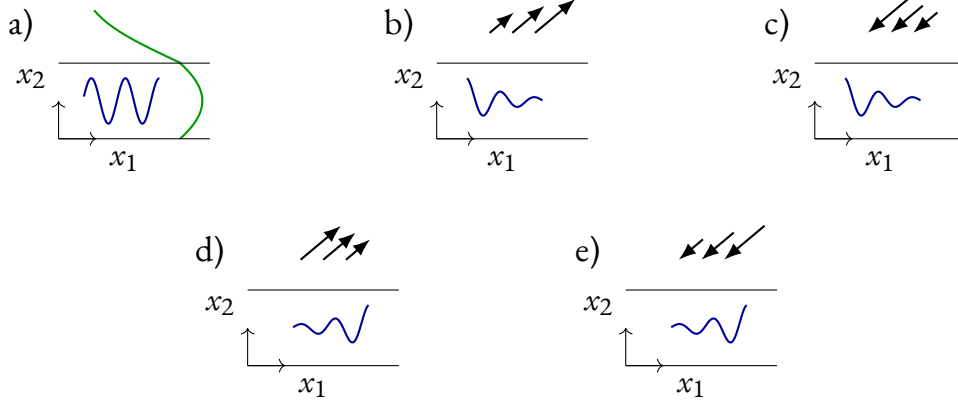


Fig. 1.6: Classes of solutions for the different types of propagating waves in open non-dissipative waveguides a) perfectly guided solution b) leaky growing waves, c) leaky attenuated waves, d) incoming attenuated waves and e) incoming growing waves.

In **conservative and closed systems**, the study is restricted to only the two wavenumber components in the layer, since no half-space is accounted. The total acoustic energy is thus confined to the structure enclosed by its boundaries. A type of wave solution exists as *perfectly guided*, this propagating wave has $k'_1 > 0$ with no attenuation $k''_1 = 0$. Solutions include *purely evanescent* wave as well $k''_1 > 0$ which often exists and continues a propagating mode below its cut-off frequency as in simple waveguides configurations as presented in Section 1.3.

Open systems support the existence of the previous types of modes with additional phenomena. Since part of the energy is susceptible to radiate out from the structure, the energy is no longer confined to the layer. Some of the previously purely guided modes can thus become attenuated, $k'_1 < 0$. Waves are still *perfectly guided* with conditions $k'_{20} = 0$ with evanescent waves in the fluid $k''_{20} < 0$, meaning that waves are confined to the proximity of the layer interface as shown in Fig. 1.6 with the amplitude along the x_1 direction in blue and the amplitude along the x_2 direction. The additional condition $k'_1 = 0$ correspond to *trapped modes*, which are a set of discrete eigenvalues of the system. They can exist in waveguides where the system is confined in one direction at least, the energy can thus be localized around some obstacle or defect by the evanescent waves^{109,110}. In open systems, they correspond to waves that are evanescent at infinity. However, in open waveguides, the acoustic energy is usually able to radiate outside of the waveguide, leading to two type of waves as *leaky waves* in the case $k'_{20} > 0$ and *incoming waves* (also called backward leaky) with $k'_{20} < 0$. These solutions are respectively exponentially growing as it radiates $k''_{20} > 0$ (see the leaky growing wave represented in Fig. 1.6b and incoming growing wave in Fig. 1.6e) or attenuated $k''_{20} < 0$ (see the leaky attenuated wave represented in Fig. 1.6c and incoming attenuated wave in Fig. 1.6d). These at first counterintuitive results ensue from the requirements that the wave should be propagating $k'_1 > 0$ and attenuated $k''_1 < 0$ along the layer, and satisfying Eq. ???. We refer also to Section 3.2 of the

thesis of Kiefer¹¹¹ dedicated to the form of solutions for quasi-guided waves in the case of an elastic layer coupled to a semi-infinite fluid half-space.

The addition of **dissipation in open systems** leads to the mixing of several loss mechanisms: the leakage of waves and the dissipations within the guide (which can be of whatever nature). In this case, the separation between incoming and leaky waves is not possible anymore. This particular aspect has been very scarcely tackled in the literature: an unpublished paper is interested in an elastic waveguide⁹³ while in optics, a paper discusses the waves guided in nanospheres with a complex permittivity¹¹². There, the modes are sorted as *proper* and *improper* solutions. This complex case is covered in a dedicated section ?? in chapter ??.

In the next parts of this chapter, a discussion on available methods to compute the complex dispersion relation for guided waves in single or stacks of homogeneous layers is proposed. There are semi-analytical methods that can be used, Section 1.3.2 or numerical methods that involve some sort of discretization in Section 1.3.3.

1.3.2. Computing the modes of porous layers using semi-analytical tools

The literature on the dispersion relation of porous/poroelastic-based structures is scarce; Groby *et al.*¹¹³ have shown the complex dispersion relation in a homogeneous equivalent fluid layer coupled to air while Boeckx *et al.*⁹² have computed the modes propagating in a poroelastic waveguide coupled at each top and bottom interface with an outgoing plane wave propagating in an air medium. This has been done using an analytical description of the system by expressing the interface conditions in a linear system (which we call GM in this work). Both of these work have used a root-finding procedure in order to solve the modes of the structure.

Using Eq. (1.19), the non-trivial solution of the problem requires that $\det(\mathcal{M}(k_1, \omega)) = 0$. The determinant of the matrix from Eq. (1.18) can be further expressed as,

$$2 \frac{k_{2,eq} k_{2,0}}{\rho_{eq} \rho_0} \cos(k_{2,eq} h) + i \left(\frac{k_{2,eq}^2}{\rho_{eq}^2} + \frac{k_{2,0}^2}{\rho_0^2} \right) \sin(k_{2,eq} h) = 0. \quad (1.30)$$

This expression can not be solved analytically. Solving wavenumbers can be done through the writing and solving of a (possibly generalized) eigenvalue problem or by root-finding of solutions in the case where writing the first one appears impossible. From this expression, one can use a root-finding procedure to solve for either the term $k_{2,eq}$ or k_1 to obtain the dispersion relation by varying the frequency. Results are shown in fig. 1.7.

We reference to Appendix ?? for extensive details on the root-finding procedure based on the Müller method. These results show an infinite amount of modes that have increasing attenuation, only the first four are shown in Fig. 1.7a to Fig. 1.7c. A branch corresponding to k_{eq} is found by the root-finding procedure, as setting $k_1 = k_{eq}$ in Eq. 1.30 indeed leads to a trivial solution. Only one propagating mode exists in the frequency range shown (one branch lies between the lines k_0 and k_{eq}) as is the expected behaviour of a lossy fluid layer.

A similar configuration, with a layer modeled using the full Biot theory is shown in Fig. 1.7d to Fig. 1.7f. Results are significantly different as the dispersion relation presents a large amount of branches. Since the elastic motion of the layer is modeled, quasi-Lamb modes are obtained along with the modes of the fluid and modes originating from the coupling of the two phases. The symmetric and antisymmetric quasi-Lamb modes are referred to with the classical Lamb

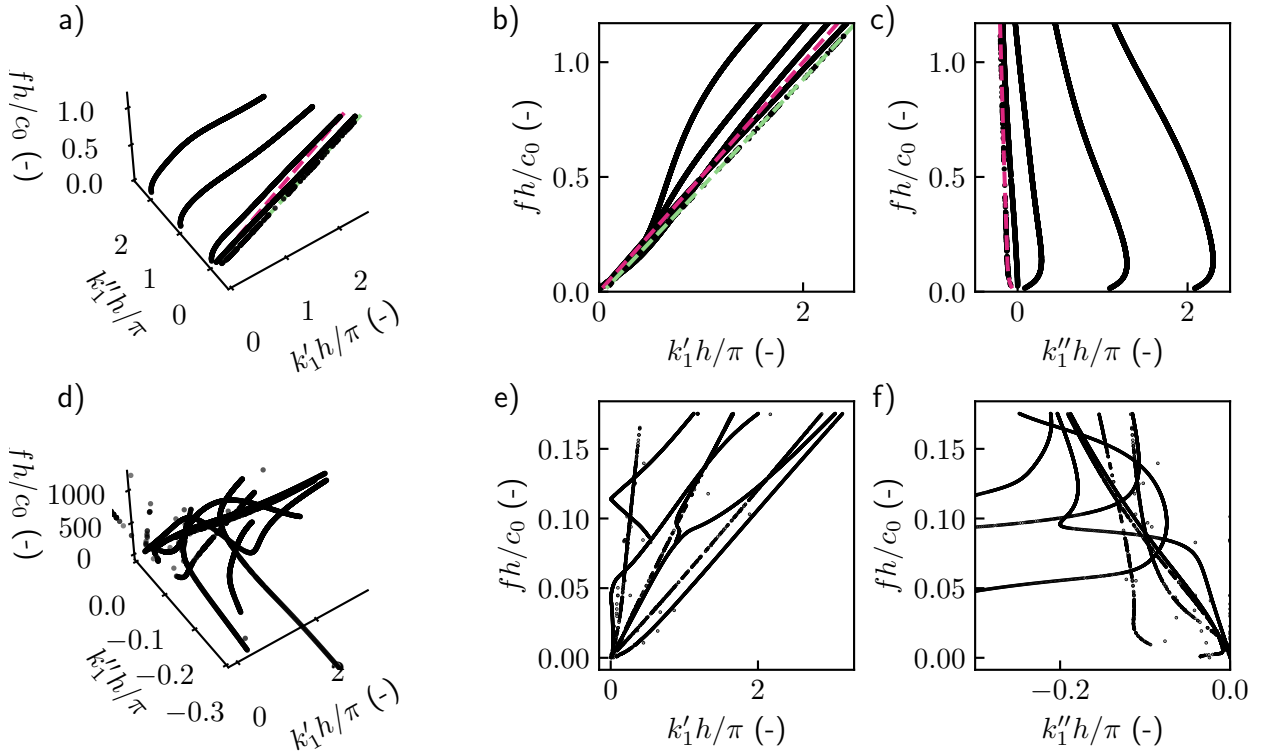


Fig. 1.7: Dispersion relation of a porous layer loaded by an air half-space on both sides. a) 3D view with k_0 (—) and k_{eq} (---), b) k_1' plane, c) k_1'' plane. d-f) poroelastic layer

mode notation the A0, S0. These results are similar to those of Boeckx *et al.*⁹². **finir quand la figure sera clean**

Generally, characteristic equations as Eq. 1.30 can only be obtained for very simple cases as is the case here, for a single fluid layer. The characteristic equation for Lamb modes is also well known¹¹⁴. However, more complicated geometry requires to solve the root-finding iterative process by using directly a numerically-computed matrix determinant which is computationally very intensive, or in many case, it is outright not possible due to the analytical complexity of the description of the system. One of the main downside of these methods is that they require multiple initial guesses to construct an interpolating function. There is no assurance of finding a solution in a given location: depending on the solution space smoothness and shape, the iterative process might get stuck in local minima, or can simply not converge in some situations. In cases where a complete dispersion relation is complicated to be constructed, more complex numerical method could thus be required; a discretization of the geometry and an interpolation of the physical fields is done as will be discussed in the upcoming section.

1.3.3. Numerics for modes computation

Several numerical methods have been investigated in the last decades with the goal of computing complex modes in homogeneous layers. Guided waves are of interest in a wide range of geometries and we focus here on the case of unbounded single (or stack of) layers with two infinite interfaces as tackled in the previous sections.

Even if a semi-analytical description of the system is possible, it can be advantageous to be able to write the problem differently, as root-finding routines often lead to challenges

as discussed in earlier sections. With that in mind, several numerical methods have been introduced with the same framework: a 1D discretization of the layer along the thickness and assuming harmonic motion along the longitudinal direction.

Modeling of the semi-infinite radiating space

There is a discussion to be made on the relevance of the strategy to use for the radiating medium modelling. In the literature, two have been presented: either the coupling can be imposed locally at the interface condition, or the domain of the surrounding medium is fully modeled, accounting for some termination condition satisfying a Sommerfeld radiation condition, by using a Perfectly Matched Layer (PML) or some analogous absorbing layer. The use of artificial absorbing layers and the discretization at the end leads to the apparition of a continuum of modes of the cavities¹¹⁵ which requires to be sorted out. A strategy based on a local coupling between the structure and the radiating medium leads to impose a form of solution for the radiation, usually as plane waves¹¹⁶. An alternative for local coupling has been investigated by associating a Boundary Elements Method (BEM) formulation for the interface¹¹⁷.

We argue that the advantages of the imposition of a local coupling is two-fold: it requires less degrees of freedom for any sort of discretization since the interface condition describes the full motion of the radiated waves. For dispersion problems, this is particularly interesting since leaky mode solutions are thus directly obtained, without needing to filter out all radiation modes linked to the additional cavity and the resonances of the PML. Also, as explained in Section ??, leaky waves are characteristic solutions of open waveguides that show exponential growth as they radiate away from the layer, which is especially ill-suited for implementing PMLs for this kind of system, although it has been achieved^{115,118}. We again refer to the thesis of Kiefer¹¹¹, especially section 3.1, for a in-depth discussion of modeling the full infinite medium against only an interface condition coupling with the layer.

Numerical methods used in the literature

Spectral methods have been around for quite some time and have been used for many numerical applications. We focus for this work on Spectral Collocation Method (SCM)¹¹⁹, which has been especially used for the computation of guided wave modes in elastic layers. One of the first paper that introduced this method for the computation of complex dispersion relations is that of Adamou *et al.*¹²⁰, while other have focused on multilayer anisotropic structures¹²¹ or cylinders¹²². These references propose an implementation for closed systems only. We refer to the introduction of Chapter 2 for an extensive bibliography on SCM applications.

This method is useful for the discretization of 1D geometries as is the object of the present section. In the case of simple 2D geometries, a 1D discretization along the transverse direction associated to a harmonic motion in the longitudinal direction. Even if the use of finite elements seems farfetched, this method could be relevant in contexts where boundary effects or guide with varying cross-sections are studied. However, a variation of finite elements based on a 1D discretization of the waveguide is entitled Semi-analytical Finite Elements (SAFE) and is akin to SCM which we have described in the previous paragraph. The 1D discretization is done along the thickness of the layer by regular nodes, and the fields are interpolated using finite elements shape functions. Examples of Semi-analytical Finite Elements (SAFE) use cases are very similar to SCM and include. A detailed description of the calculation of Lamb modes using a simple

SAFE implementation using high-order Lobatto polynomials is proposed in Appendix ???. There, its performance is also compared to that of SCM. The particular advantage of SCM is the very quick convergence allowed by the interpolation using Chebyshev polynomials, called *spectral convergence* in the literature^{119,123,124}. As it was shown in the aforementioned appendix, we saw that it outperforms higher-order SAFE. Thus, we will focus on this method for the modeling of multilayer homogeneous structures.

Open elastic layers accounting for surrounding air media that couples the elastic layer have been first presented by Kiefer *et al.*¹¹⁶ where a SCM was used to discretize the elastic layer. Additionally, a local coupling is modeled with a discretized interface conditions. The system of equations is then formulated as a GEP and can be solved with traditional eigenvalue solvers. This work forms the basis of the work done in Chapter 2. Other implementations of the same configuration include a discretization of the fluid half-space by means of a mapping allowing the numerical solution to increase exponentially at infinity, as is a characteristic of leaky waves¹²⁵.

With Semi-analytical Finite Elements (SAFE), which has been widely used for similar configurations as well have computed modes of immersed waveguides¹¹⁸, leaky Lamb modes using PMLs and a coupling between SAFE and BEM¹¹⁷. A detailed overview on the literature dealing with this topic is proposed in Appendix ???. Ducasse and Deschamps¹⁰⁶ have also formulated an eigenvalue problem by discretizing the transverse direction of the layer using finite differences.

1.4 | Methods for modelling metaporous surfaces and interfaces

1.4.1. A word on periodic structures

A large literature that shows the enhancement of absorption properties by periodicizing porous layers with scatterers was discussed in the introduction. In this section, a presentation of the fundamental notions surrounding periodic structures is proposed. In wave physics, periodic structures have been shown to introduce interesting effects towards applications for wave control and constitute the cornerstone of acoustic metamaterials. The periodicity can be imposed in acoustic systems using the so-called Bloch-Floquet theorem. This formalism has been pioneered by Felix Bloch and Léon Brillouin among others^{86,126}. Throughout this manuscript, we consider a periodic cell along only the longitudinal direction, that 1D periodicity which we denote d . Taking an infinitely repeating lattice or a crystal, the description of the wave propagation in only one of the so-called unit cell, one spatial period of the total structure, is enough to describe the whole structure. The layer of infinite extent is thus limited to a single unit-cell. A wavefunction $\psi(x)$ describing the propagation should thus depend on a field with the same spatial periodicity as the lattice. To do so, the wavefunction is projected onto an infinite modal basis, that is Bloch modes,

$$\psi_q(x) = e^{ik_q x_1}. \quad (1.31)$$

The wavefunction at the left boundary is thus related to that at the right boundary modulated by the spatial periodicity and some translation vector, that is $\psi(x + qd) = e^{ik} \psi(x)$.

The possibility of reduction of the spatial considerations of the problem also leads to a reduction of the reciprocal wavenumber space. In periodic structure, we often restrict this reciprocal space to the first Brillouin zone, that is to wavenumbers restricted to $k_{1q} \in [0, \pi/d]$ when looking at the band structure. The consequences of periodic media are directly observable in the band structure: bands are usually folded when the wavenumber reaches this limit of the reciprocal space. This folding occurs at the Bragg frequency corresponds to a relation between the wavelength λ of the excitation signal and the periodicity of the crystal lattices, $d \sin(\theta) = q\lambda$, with θ the angle of incidence. This condition corresponds to destructive interferences: this phenomenon is a cause of the appearance of the famous Bragg bandgaps; in these bands of frequency, waves are unable to propagate. Variations of these phenomenon have been investigated in acoustic metamaterials: e.g. the addition of periodic resonators leads to hybridization stopbands. [ref these](#)

1.4.2. Methods to compute the scattering of waves by periodic structures

[manque des refs partout](#) Analytical methods for the evaluation of the wave propagation in periodic structures often require tedious derivations, as the frequency regimes of interest is usually high enough to require the description of the multiple scattering happening in the structure. The scatterer of the unit cell onto which the incident wave impinges acts as a secondary source, which scattered field is successively scattered by each neighboring obstacles and so on. Multiple Scattering Theory is a theoretical framework that has been developed to account for these phenomena and was used to compute the acoustic response of metaporous structures with rigid inclusions⁷¹ and metaporoelastic surfaces with solid and shell inclusions⁸². Both configurations have been tuned to show a perfect absorption peak in the frequency spectrum.

The remaining papers discussing how incident pressure fields are affected and scattered by metaporous structures use FEM, for the most part through the interface of Comsol Multiphysics. This very versatile and powerful method has been used for decades in all fields of physics and thus won't be detailed in this thesis. It is recommended to the reader as an introduction on the basis of FEM, the textbooks from Atalla and Sgard¹²⁷ related to acoustics and vibration problems and that of Dhatt *et al.*¹²⁸ and Solin *et al.*¹²⁹ dealing specifically with higher-order finite elements problems. This method allows for the discretization of the geometry via a mesh in order to solve the appropriate motion equations. This has allowed to use rather complex geometries, e.g spirals¹³⁰ and labyrinthine paths⁶⁸ as scatterers embedded in porous layers as discussed in the state of the art. Additionally, Gaborit *et al.*⁸³ have developed a method in order to couple FEM with Bloch modes expansion of analytical fields described by TMM. They have used it to compute the scattering coefficients of structure that include a periodic cell sandwiched by homogeneous layers with an embedded inclusion.

1.4.3. Finite elements

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- reprendre sur comment arriver à NME eq.22: formulation eq. faible + maillage + interpolation polynomiale (higher order) aux nodes du mesh

- sert à étoffer la partie du chapitre 4: écrire la recette permettant d’y arriver.
- cas du fluide et montrer les différentes variational formulations

1.5 | Dispersion relations and band structure in periodic lattices

1.5.1. Methods for non-radiative systems

One of the most popular methods for the computation of band structures in periodic systems made of classical lossless materials are the Plane Wave Expansion (PWE) which is semi-analytical. It relies on a Fourier transform of the motion equations, which allows an expression of the fields expanded onto Bloch modes, truncated for numerical applications. This method, while easy to implement presents important limitations due to the convergence of Fourier series and the matching of constituent materials for the lattice: materials should preferably be of the same nature for the calculation to go well²⁷. Generalized as the Extended Plane Wave Expansion (EPWE), the method also allows to obtain the evanescent Bloch modes in Ref.¹³¹, allowing for complex dispersion diagram investigation and to account for evanescent modes in finite 2D periodic lattices.

The complexity of the system prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods. As the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation. The thesis of Magliacano^{magliacano2020a} was focused in the modeling of a closed system, of an equivalent fluid unit cell with an embedded rigid inclusion. The finite elements formalism was entitled with the so-called shift cell technique, that implements the Bloch-Floquet phase shift at the boundaries inside of the constitutive variational weak formulation.

1.5.2. Unpromising adaptation of existing methods

The complexity of the system of interest in this thesis as presented in Section?? of the Introduction prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods: as the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

Multiple scattering could also be considered. However, it has been successfully used to describe the acoustic scattering of periodically arranged rigid inclusions in equivalent fluid unit cells⁷¹ as well as solid inclusions in a poroelastic unit cell⁸². However, the so-called Schlömilch series that appear in the calculation, from the effect of the scattering of the neighboring cylindrical inclusions appear to be not so manipulable when considering complex longitudinal

wavenumbers. Its expression is

$$\sigma_q^X = \sum_{l=1}^{\infty} H_q^{(1)}(k^X l d) \left((-1)^q e^{i k_1^X l d} + e^{-i k_1^X l d} \right), \quad (1.32)$$

and indeed, separating the k_1^X real and imaginary inevitably leads to at least one growing exponential term in the series, leading to it being unable to converge, up to our study.

We face challenges when trying to model this structure using the tools classically used in the literature, that is COMSOL Multiphysics. Although there are built-in tools in the software for "Eigenfrequency" studies, the library aims at solving complex frequencies of the system, given a set of parameters for the wavenumber. As it has been discussed earlier in the chapter, the goal of this work is to solve for complex wavenumbers and real frequencies. Even trying a manual implementation of the model in the "Weak Form PDE" interface, the tendency of this software to function as a black box introduces a new degree of complexity to this strategy. While FEM remains the soundest strategy for the modeling of the complex inhomogeneous geometry of the unit cell of metaporous surface, we propose to adapt in-house existing frameworks in order to develop numerical tools in this thesis.

1.6 Conclusion and prospects of methods for the thesis

In the case of homogeneous multilayer systems with surrounding fluid loading, we have seen that semi-analytical have long been established in the literature in order to compute their scattering properties. For dispersion problems however, these configurations usually require some sort of numerical discretization to treat the geometry; a wide range of methods allowing for the direct formulation of an eigenvalue problem and a computation of modes was shown in Section 1.3 with first semi-analytical tools applicable to simple cases (e.g. an equivalent fluid layer with surrounding air), and then various numerical methods like SCM or SAFE as the two main protagonist.

When going to periodic systems, it appeared that the way to tackle the problem is quite different. Indeed, the resulting complex geometry (due to the obstacles) leads to much different tools. In the case of scattering properties computation, semi-analytical tools are still available. We have seen that FEM seems to provide promising possibilities for the modelling of complex periodic cores. However, the considerations of open systems leads to question regarding the modeling of the half-spaces surrounding the structure. Two strategies compete in the literature: either a fluid cavity is added to the geometry and terminated by some sort of absorbing layer at the end; or the harmonic fluid radiation is locally accounted for as an analytical interface condition of the structure. **parler de couplage avec plaque et sandwich**

A general spectral collocation method for computing the dispersion relations of guided acoustic waves in multilayer dissipative structures

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THIS CHAPTER IS BASED ON THE MATERIAL PUBLISHED IN^{P1}, AUTHORED BY *MATHEU MARÉCHAL, ALAN GESLAIN, JEAN-PHILIPPE GROBY, VICENT ROMERO-GARCÍA AND OLIVIER DAZEL*. MINOR MODIFICATIONS TO THE TEXT AND FIGURES HAVE BEEN MADE TO FIT THE FORMAT AND NOTATION OF THIS THESIS⁽¹⁾

⁽¹⁾While the remaining of the manuscript assumes a positive harmonic time-dependance, the reader should bear in mind that this entire chapter is written using a negative time-dependance, i.e. $e^{-i\omega t}$.

2.1 | Foreword

There are fundamental differences between this chapter and the rest of the manuscript in how the leakage is treated in the approach. In this chapter, no questions were raised regarding the wave solutions in the surrounding media. As a first assumption, we considered that the leaky waves should be outgoing in order to satisfy some radiation condition as stated right before Eq. 2.5. However, this matter is not so simple, and treated as part of the subsequent chapter. In that respect, the dispersion relation of the waves in this chapter show indiscriminately outgoing as well as incoming solution. Fortunately, in these configurations the solution spectrum in real parts of the transverse wavenumber are almost symmetrical for most cases, which is why we did not consider this aspect in this chapter.

Therefore, contrary to the indication given below Eq. 2.5 giving the expressions the pressure fields in the surrounding media, it is stated in the original article that the branch cut for the transverse wavenumber in the half-space is selected such that $\text{Re}(k_2^{(0)}) \geq 0$. As discussed in the above paragraph, the eigenvalue problem uncovers all solution without a requirement for a branch cut selection thanks to the γ variable change.

2.2 | Abstract

A spectral collocation method is proposed to compute the complex wavenumber-real frequency dispersion relations of guided acoustic waves in multilayer structures involving dissipative materials. The nature of these dissipative materials is initially considered to be arbitrary, i.e., poroelastic, viscoelastic, or viscoacoustic. For a given frequency, the complex wavenumbers as well as the physical fields, which are further used to evaluate the Poynting vectors and analyze the energy flux, are obtained by solving a generalized eigenvalue problem. The latter arises from a set of discretized equations of motion and appropriate boundary (coupling) conditions. These equations of motion and boundary (coupling) conditions are imposed by the nature of the material composing each layer of the structure. A focus is made on poroelastic layers. The dispersion relation of a two-layer elastic-poroelastic structure is analyzed, as well as the energy flows in the structure. The results as calculated with the present spectral collocation method are validated against those obtained with a classical complex root-finding (Müller) method and experiments.

2.3 | Introduction

Dealing with guided acoustic waves in multilayer structures of dissipative (viscoelastic, viscothermal, or poroelastic) material requires the evaluation of complex wavenumber-real frequency dispersion relations. Solving for these dispersion relations consists in finding complex roots of complex characteristic equations, which often arise from complex matrix determinants and can rarely be solved analytically. Various methods^{132,133}, among which the Müller algorithm¹³⁴, are commonly used to achieve this challenging task. These methods usually rely

on preliminary semi-analytical derivations, require multiple initial guesses, and are iterative, so they can get stuck in local minima. Computing the complex wavenumber-real frequency dispersion relation in a reliable and robust way is thus of primary importance. Commercial software such as Disperse¹³⁵ or open-source software such as Dispersion Calculator^{136,137} are available for calculating dispersion curves. They use mode tracing or root-finding methods and focus mainly on multilayer elastic structures. Alternatively, several numerical methods can be used, such as the Finite Element Method¹⁰⁸, the Semi-Analytical Finite Element method (SAFE)^{138–140}, or spectral methods¹²³. Spectral methods are especially relevant in this framework. They are numerical approximation techniques that seek the solution of a differential equation using a finite series of infinitely differentiable basis functions with *spectral accuracy*¹¹⁹.

The propagation and dispersion of waves in multilayer systems have long been investigated^{102,103}. Spectral methods have been developed to cope with wave radiation problems, but have also emerged in the last decades to solve dispersion relations¹⁴¹. Effectively, the numerical scheme boils down to discretizing the geometry along the layer thickness and solving an eigenvalue problem. Here, a collocation method is used to discretize the geometry. A finite-dimension polynomial basis is chosen and the collocation points are distributed according to the roots of this polynomial basis. Leaky modes appear in practice when multilayer structures are surrounded by one or two half-spaces. Accounting for this phenomenon brings us closer to practical considerations regarding these systems, as they would radiate part of the acoustic energy back to the surrounding medium, although it adds hardships to the modeling of the system; the waves modeled in these half-spaces should satisfy some radiation condition. These can be imposed by implementing perfectly matched layers¹¹⁵, by using a bounded mapping, and applying a Spectral Collocation Method (SCM) to all layers of the extended system¹²⁵, or by using a suitable change of variable and solving a non-linear eigenvalue problem¹¹⁶. This last method inspires the current work. We propose a SCM suitable for solving complex wavenumber-real frequency dispersion relations in multilayer systems involving dissipative media. As a way of example, we will focus on multilayer poroelastic systems. Mechanical wave propagation in bulk poroelastic materials are well understood and modeled thanks to the seminal contributions of M.A. Biot^{98,99,142,143} and subsequent contributions, notably by D.L. Johnson *et al.*⁹⁴ and J.-F. Allard *et al.*^{19,95} concerning the modeling of viscothermal losses. Although poroelastic materials are bounded in practice and are usually encapsulated in more complex multilayer structures combining layers of materials of different nature in geophysics and acoustics, there are only a few studies on the dispersion relation of guided acoustic waves in poroelastic^{82,92,144}, porous^{groby2008a, 145} or anisotropic poroelastic layers¹⁴⁶. The SCM has already been used to solve the dispersion relation of guided acoustic waves in elastic layered rings^{120,122} or anisotropic elastic multilayer systems¹²¹, but has never been used to cope with multilayer systems involving dissipative materials of arbitrary nature.

This work aims at filling this gap and is divided into two sections. First, the SCM is derived for generic multilayer configurations possibly coupled to two surrounding identical fluid half-spaces. A two-layer elastic-poroelastic structure is then considered by way of example. The dispersion relation computed with the SCM method is compared to that obtained with a classical root-finding (Müller) method and that measured. The energy flow is further analyzed for this two-layer configuration, with very little additional computational cost. In the last section, we experimentally analyze the two-layer structure by retrieving its complex dispersion

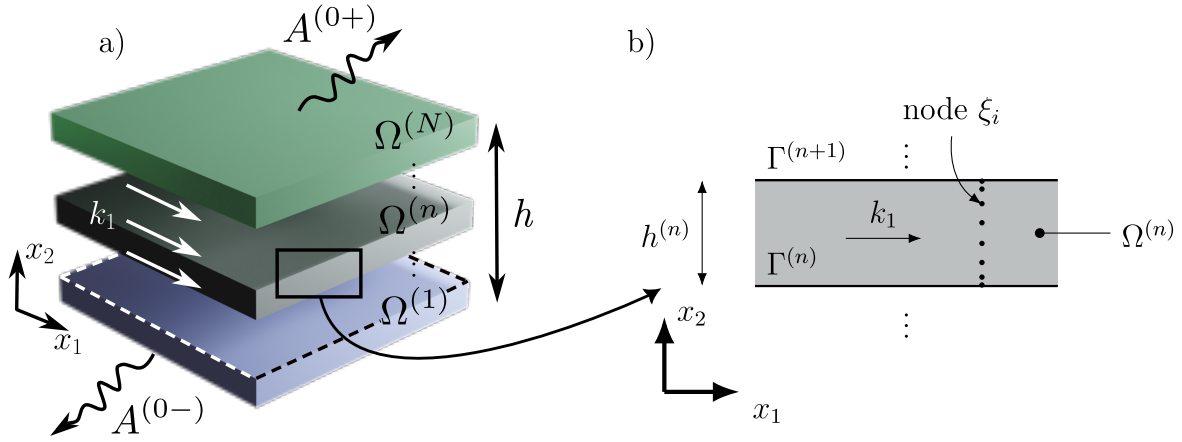


Fig. 2.1: a) Sketch of the geometry of the multilayer system and b) zoom on the n -th layer.

relation from displacement measurements using the SLaTCoW (Spatial Laplace Transform for COMplex Wavenumber recovery). The previously obtained numerical results can thus be correlated with the data of the dispersion relation.

2.4 Solving the dispersion relations for multilayer structures

2.4.1. Spectral Collocation Method

Figure 2.1a depicts the general geometry of the multilayer structure considered in this article. It is composed of N homogeneous layers of arbitrary nature. The n -th layer has a thickness of $h^{(n)}$ and occupies the domain $\Omega^{(n)}$. The lower and upper interfaces are denoted $\Gamma^{(n-1)}$ and $\Gamma^{(n)}$. The multilayer structure is assumed to be two-dimensional, of infinite extent along x_1 , and bounded along the x_2 -axis between 0 and the total structure thickness $h = \sum_{n=1}^N h^{(n)}$. Outgoing waves radiate above and below the structure in two fluid half-spaces $\Omega^{(0\pm)}$, occupied by the same fluid.

We assume an implicit time dependence $e^{-i\omega t}$. The physical fields associated to the guided waves in the n -th layer propagating along the positive x_1 -axis take the form

$$\Theta^{(n)}(x) = s^{(n)}(k_1, x_2)e^{ik_1x_1}, \quad (2.1)$$

where $s^{(n)}(k_1, x_2)$ is the spatial Laplace transform of $\Theta^{(n)}(x)$ and k_1 is the complex valued x_1 -component of the wavenumber. Note that the spatial Laplace transform is used instead of the spatial Fourier transform because k_1 is complex-valued. The physical fields that compose $\Theta^{(n)}(x)$ depend on the nature of the material occupying the n -th layer. Introducing $\Theta^{(n)}(x)$ in the corresponding wave equations (of second order in space in this work) reduces to a second-order differential operator $\mathcal{L}^{(n)}$ in x_2 , that depends on the material properties and can further be developed as a

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second-order polynomial in k_1

$$\mathcal{L}^{(n)}(x_2) s^{(n)}(k_1, x_2) = \left(k_1^2 \mathcal{L}_2^{(n)}(x_2) + k_1 \mathcal{L}_1^{(n)}(x_2) + \mathcal{L}_0^{(n)}(x_2) \right) s^{(n)}(k_1, x_2) = \mathbf{0}, \quad (2.2)$$

where $\mathcal{L}_j^{(n)}$ are the coefficients of the polynomial expansion $j = 0, 1, 2$. Please note that this operator is nothing but the spatial Laplace transform of the equations of motion.

The interface (boundary) conditions at $\Gamma^{(n)}$ between the $n + 1$ -th and the n -th layers again depend on the nature of these two layers, but can be formally written in the form

$$C^{(n)-} s^{(n)}(k_1) \Big|_{\Gamma^{(n)}} - C^{(n)+} s^{(n+1)}(k_1) \Big|_{\Gamma^{(n)}} = \mathbf{0}, \quad (2.3)$$

where $C^{(n)\pm}$ are interfaces operators, which involve differential operators of maximum second-order in x_2 . Following this procedure and introducing $\mathcal{S} = \left(s^{(0-)} \dots s^{(n)} \dots s^{(0+)} \right)^T$, with T denoting transposition, the problem can be cast in matrix form, whose coefficients are differential operators in x_2 ,

$$K S(k_1, x_2) = \begin{pmatrix} C^{(0-)} & -C^{(0+)} & 0 & & & \\ 0 & \mathcal{L}^{(1)}(x_2) & 0 & & & \\ 0 & C^{(1-)} & -C^{(1+)} & 0 & & \\ & 0 & \mathcal{L}^{(2)}(x_2) & 0 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 0 & \mathcal{L}^{(N)}(x_2) & 0 \\ & & & 0 & C^{(N-)} & -C^{(N+)} \end{pmatrix} S(k_1, x_2) = \mathbf{0}. \quad (2.4)$$

Note that the equations of motion in both upper and lower half-spaces are not required in Eq.(2.4). This system only depends on the values of the two spatial Laplace transforms $s^{(0\pm)}$ at the upper and lower interfaces. Instead, we specify the nature of the two half-spaces, i.e. a fluid medium, and make use of the Sommerfeld condition, to explicitly give the form of the spatial Laplace transforms $\tilde{p}^{(0\pm)}(k_1, x_2)$ of the pressure fields $p^{(0\pm)}(x)$

$$\begin{aligned} \tilde{p}^{(0-)}(k_1, x_2) &= A^{(0-)} e^{-ik_2^{(0)} x_2}, \in \Omega^{(0-)}, \\ \tilde{p}^{(0+)}(k_1, x_2) &= A^{(0+)} e^{ik_2^{(0)} (x_2 - b)}, \in \Omega^{(0+)}, \end{aligned} \quad (2.5)$$

with $k_2^{(0)} = \sqrt{(k^{(0)})^2 - k_1^2}$, such that $\text{Re}(k_2^{(0)}) \in \mathbb{R}$, where $k^{(0)} = \omega/c_f$ is the wavenumber in the fluid and c_f the wave velocity. ⁽²⁾ Thus, $s^{(0\pm)}$ appearing in Eq.(2.4) reduces to $s^{(0\pm)} = A^{(0\pm)}$. $C^{(0-)}$ and $C^{(N+)}$ are therefore no longer operators and reduce to $C^{(0-)} = C^{(0-)}$ and $C^{(N+)} = C^{(N+)}$. In our fluid case, the interface conditions at $\Gamma^{(0)}$ and $\Gamma^{(N)}$ involve the spatial Laplace transform of the pressure and its first spatial derivative with respect to x_2 whatever the nature of the 1-st and N -th layers. Both matrices $C^{(j)}$, $j = (0-), (N+)$, can thus be formally written as

$$C^{(j)} = k_1 C_1^{(j)} + C_0^{(j)} + k_2^{(0)} C_{1'}^{(j)}, \quad (2.6)$$

⁽²⁾The condition on the transverse wavenumber was written as $\text{Re}(k_2^{(0)}) \geq 0$ in the original article, see the foreword in Section 2.1.

with the subscript denoting the submatrices rearranged according to the power order $k1$. Index $1'$ corresponds to the term in $k_2^{(0)}$. Introducing Eqs.(2.2) and (2.6) in Eq. (2.4) leads to

$$\left(k_1^2 K_2 + k_1 K_1 + K_0 + i k_2^{(0)} K_{1'}\right) S = 0. \quad (2.7)$$

This eigenvalue problem is non-linear because of the presence of the term $k_2^{(0)}$. Making use of the following changes of variable $k_1 = k^{(0)} (\gamma + \gamma^{-1}) / 2$ and $k_2^{(0)} = k^{(0)} (\gamma - \gamma^{-1}) / 2i$ ^{147,148} and companion linearization¹⁴⁹, a generalized eigenvalue problem is formed and solved for γ , from which the dispersion relation can be calculated. More details on this procedure can be found in¹¹⁶. Note that the first or the last column of Eq.(2.4) vanishes when radiation is absent on either side of the multilayer system. The situation is different when both fluid half-spaces disappear. The eigenvalue problem Eq.(2.7) is no longer non-linear and the problem can directly be solved after companion linearization. Note also that fluid half-spaces are only considered for the sake of simplicity and that this procedure can be adapted to any kind of material occupying the half-spaces, as described recently¹⁵⁰. Note finally that this procedure avoids discretization of these two half-spaces.

To numerically solve the problem, the spatial Laplace transform of the physical fields in each n -th layer are expanded as

$$s^{(n)}(x_2) \approx \underline{s}^{(n)}(x_2) = \sum_{m=0}^M \alpha_m^{(n)} \underline{\psi}_m(\underline{\xi}^{(n)}), \quad (2.8)$$

where α_m are coefficients of the polynomial expansion, $\underline{\psi}_m$ are the Chebyshev polynomial of order m , and $\underline{\xi}^{(n)} = 2(x_2 - \sum_{j=1}^{n-1} b^{(j)}) / b^{(n)} - 1$, $\in [-1, 1]$. Underlined variables represent discrete vectors and double-underlined variables discrete matrices. The goal of this approximation is to discretize the differential operators in x_2 appearing in the problem. To do so, the SCM is employed. $\underline{\xi}^{(n)}$ is discretized on the roots of the Chebyshev polynomials, $\underline{\xi}_j^{(n)} = \cos\left(\frac{j\pi}{M^{(n)}}\right)$ with $j = 0 \dots M^{(n)}$. Please note that $\underline{\xi}_j^{(n)}$ discretely runs from 1 to -1 with increasing j , while $\underline{\xi}^{(n)}$ continuously runs over the same interval but in the opposite direction, i.e. from -1 to 1. These collocation points/nodes form a non-uniform grid along the thickness of each layer. This discretization is usually preferred because the collocation points are clustered at the interfaces¹¹⁹. The locations of these nodes are presented in Fig. 2.1b for the n -th layer with an arbitrary value $M^{(n)} = 8$. The differential operators in the n -th layer are thus represented by differentiation matrices (DMs), such that

$$\begin{aligned} s^{(n)}(x_2) &\rightarrow \underline{I} \underline{s}^{(n)}(\underline{\xi}_j), & \partial_2 s^{(n)}(x_2) &\rightarrow \left(\frac{2}{b^{(n)}}\right) \underline{\underline{D}}_2 \underline{s}^{(n)}(\underline{\xi}_j) = \underline{\underline{D}}_2^{(n)} \underline{s}^{(n)}(\underline{\xi}_j), \\ \partial_{22} s^{(n)}(x_2) &\rightarrow \left(\frac{2}{b^{(n)}}\right)^2 \underline{\underline{D}}_{22} \underline{s}^{(n)}(\underline{\xi}_j) = \underline{\underline{D}}_{22}^{(n)} \underline{s}^{(n)}(\underline{\xi}_j). \end{aligned} \quad (2.9)$$

where $\underline{I}$ is the identity matrix and $\underline{\underline{D}}_2$ and $\underline{\underline{D}}_{22}$ are respectively the first-order and second-order normalized DMs. Contrary to an interpolation with Chebyshev polynomials, DMs interpolate

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the functions and their spatial derivatives directly and thus do not involve the polynomial expansion coefficients per se.¹⁵¹ Note that the normalized DMs are denoted as $\underline{\underline{D}}_2^{(n)}$ and $\underline{\underline{D}}_{22}^{(n)}$ in the following. These matrices are square and have a size of $(M \times M)$. This size also directly depends on the order of the considered Chebyshev polynomials. The formulation prevents the occurrence of roundoff errors, which is of particular importance when SCM is applied to solve eigenvalue problems¹⁵².

The linear operator $\mathcal{L}^{(n)}$ is discretized as a matrix $\underline{\underline{L}}^{(n)}$ with size $(P^{(n)}(M^{(n)} - 1), P^{(n)}(M^{(n)} + 1))$, where $P^{(n)}$ is the number of physical fields required to describe the wave propagation in the layer and again depends on the nature of the material this layer is composed of. The first and the last rows of the DMs are removed because they are used to apply the interface conditions at $\Gamma^{(n+1)}$ and $\Gamma^{(n)}$ respectively. The interface operators $C^{(n)\pm}$ are effectively expressed in terms of this first row of the DMs of the n -th layer and this last row of the DMs of the $(n + 1)$ -th layer. These rows respectively correspond to the first, i.e. $\underline{\xi}_{M^{(n)}} = -1$ and $\underline{\xi}^{(n)} = 1$, and last collocation nodes of the n -th and $(n + 1)$ -th layers, i.e. $\underline{\xi}_0 = 1$ and $\underline{\xi}^{(n+1)} = -1$. These vectors are subsequently denoted $\underline{I}^{(n+1)-}$, $\underline{I}^{(n)+}$, $\underline{D}_2^{(n+1)-}$, $\underline{D}_2^{(n)+}$, $\underline{D}_{22}^{(n+1)-}$, and $\underline{D}_{22}^{(n)+}$. The interface matrix $\underline{\underline{C}}^{(n)-}$ is of size $(P^{(n)} + P^{(n+1)}, P^{(n)}(M^{(n)} + 1))$ and the interface matrix $\underline{\underline{C}}^{(n)+}$ is of size $(P^{(n)} + P^{(n+1)}, P^{(n+1)}(M^{(n+1)} + 1))$.

Finally, the full $(\sum_{n=0}^N M^{(n)} P^{(n)} + 2, \sum_{n=0}^N M^{(n)} P^{(n)} + 2)$ -matrix system for the general multilayer system is

$$\underline{\underline{K}} \underline{\underline{S}} = \begin{pmatrix} \underline{\underline{C}}^{(0-)} & -\underline{\underline{C}}^{(0+)} & 0 & & & \\ 0 & \underline{\underline{L}}^{(1)} & 0 & & & \\ 0 & \underline{\underline{C}}^{(1-)} & -\underline{\underline{C}}^{(1+)} & 0 & & \\ & 0 & \underline{\underline{L}}^{(2)} & 0 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 0 & \underline{\underline{L}}^{(N)} & 0 \\ & & & 0 & \underline{\underline{C}}^{(N-)} & -\underline{\underline{C}}^{(N+)} \end{pmatrix} \begin{pmatrix} \underline{A}^{(0-)} \\ \underline{s}^{(1)} \\ \underline{s}^{(2)} \\ \vdots \\ \underline{s}^{(N)} \\ \underline{A}^{(0+)} \end{pmatrix} = \underline{0}. \quad (2.10)$$

This system is cast in the form of a generalized eigenvalue problem that is solved by traditional eigenvalue solvers at each frequency. The dispersion relation for the acoustic guided wave in the dissipative multilayer system is calculated, repeating the procedure for each frequency ω .

2.4.2. Solving the dispersion relation with Müller algorithm

The dispersion relation of multi-layer dissipative structure are more commonly calculated with secant-based algorithm as the Müller algorithm. The latter relies on a semi-analytic description of the fields. These fields are best written in each layer, independently of the nature of the material it is composed of, using the potentials, i.e. decomposing the field $\theta = \nabla \varphi + \nabla \times \psi$ into an irrotational component φ and a transverse component $\psi = \psi e_3$. The spatial Laplace transform of the potentials in the n -th layer are entirely described by

$$\tilde{\theta}_n(k_1, x_2) = A_X^{(n)+} e^{ik_{2,X}^{(n)}(x_2 - b^{(n)})} + A_X^{(n)-} e^{-ik_{2,X}^{(n)}(x_2 - b^{(n)})}, \quad (2.11)$$

with $A_X^{(n)\pm}$ the up and down-going wave amplitudes, $k_{2,X}^{(n)} = \sqrt{(k_X^{(n)})^2 - k_1^2}$, such that $\text{Re}(k_{2,X}^{(n)}) \geq 0$,⁽³⁾ and X refers to the type of wave, i.e. shear, compressional or acoustic. If necessary, these Laplace transform representations are complemented by those of the pressure field in the fluid half-spaces provided in Eq.(2.5).

The boundary conditions are applied at each interface and the set of equations is cast into the following matrix form $\mathbf{M}(k_1)\mathbf{q} = 0$, with $\mathbf{q}$ a vector containing the amplitudes of each field/potential. The modes of the system correspond to the wavenumbers k_1 for which the matrix $\mathbf{M}(k_1)$ is singular, i.e.

$$\det(\mathbf{M}(k_1)) = 0. \quad (2.12)$$

These wavenumbers are the complex roots of an implicit complex-valued equation. They are evaluated by running our own implementation of the Müller algorithm¹³⁴ from 3 initial guesses. This equation is solved iteratively for each frequency ω , giving the dispersion relation.

In addition, $\mathbf{q}$ can be obtained by computing the nullspace of $\mathbf{M}$ ¹⁵³ (This null space is computed using singular value decomposition as embedded in the `linalg.null_space` function from the Python package SciPy). The amplitudes of the potentials in the layers are thus evaluated, enabling all the physical fields that depend on these potentials to be calculated. However, the $\mathbf{M}$ matrix becomes increasingly tedious to write as the number of layers increases, and even more so when it comes to calculating the roots of its determinant and nullspaces. Nevertheless, this approach remains a fairly effective way of providing a reference solution for validating numerical results when studying a two-layer system.

2.5 Application to a two-layer poroelastic-elastic structure

In this section, the calculation of the dispersion relation for guided waves in a two-layer structure, i.e. $N = 2$, is used as an example of the application of the present SCM. This two-layer structure consists of a $b^{(1)}$ -thick poroelastic layer, coated on one side by a $b^{(2)}$ -thick aluminum plate, that is radiating in a fluid half-space, and rigidly backed on the other side as depicted in Fig. 2.2a. The SCM results are analyzed and compared to those obtained with the Müller root-finding method.

2.5.1. Numerical scheme

The pressure field $p^{(1)}$ and the two components of the frame displacement, $u_{1,s}^{(1)}$ and $u_{2,s}^{(1)}$, i.e. $P^{(1)} = 3$, as well as the two-components of the elastic displacement, $u_1^{(2)}$ and $u_2^{(2)}$, i.e. $P^{(2)} = 2$, are used to model the propagation in the poroelastic and in the elastic layers respectively. The $\{\mathbf{u}_s, p\}$ formulation is used¹⁰⁰ for the poroelastic medium. The more usual displacement formulations ($\{\mathbf{u}_s, \mathbf{u}_f\}$ ⁹⁸ or $\{\mathbf{u}_s, \mathbf{w}\}$ ¹⁴²) cannot be used in the present case, because the number of physical fields they rely on, i.e. 4, is different from that of the potentials in the layer, i.e. 3.

⁽³⁾Following the discussion from the foreword in Section 2.1, the square roots in this part do require a branch cut selection, meaning here that outgoing solution are considered.

2.5. Application to a two-layer poroelastic-elastic structure

In total, 5 physical fields inside the two-layer structure and an additional amplitude for the acoustic wave radiated towards $x_2 \rightarrow \infty$ are needed to solve for the dispersion relation. Each domain is discretized on $M^{(j)}$, $j = 1, 2$, collocation points.

In more details, the $\{\mathbf{u}_s, p\}$ formulation of the Biot theory¹⁰⁰ is employed to model the propagation in the homogeneous isotropic poroelastic layer. The coupled equations of motion read as

$$\begin{cases} \nabla \cdot \hat{\boldsymbol{\sigma}}_s^{(1)} + \hat{\rho}\omega^2 \mathbf{u}_s^{(1)} + \beta \nabla p^{(1)} = 0, \\ \frac{\nabla^2 p^{(1)}}{\rho_{22}\omega^2} - \frac{\beta}{\phi^2} \nabla \cdot \mathbf{u}_s^{(1)} + \frac{1}{R} p^{(1)} = 0, \end{cases} \quad (2.13)$$

where $\hat{\boldsymbol{\sigma}}_s^{(1)} = \hat{A} \nabla \cdot \mathbf{u}_s^{(1)} \mathbf{I} + 2N_s \boldsymbol{\epsilon}_s$ is the *in vacuo* solid stress tensor, with $\boldsymbol{\epsilon}_s$, the solid-phase strain tensor. The parameters are generally expressed in terms of the elastic coefficients P , Q and R and the effective densities ρ_{11} , ρ_{12} and ρ_{22} initially introduced by Biot^{11,98} as

$$\hat{A} = P - 2N_s - \frac{Q^2}{R}, \quad \hat{\rho} = \rho_{11} - \frac{\rho_{12}^2}{\rho_{22}}, \quad \beta = \phi \left(\frac{\rho_{12}}{\rho_{22}} - \frac{Q}{R} \right). \quad (2.14)$$

These parameters depend on the frame properties and on the effective density $\rho_{eq}(\omega)$ and bulk modulus $K_{eq}(\omega)$ of the fluid phase as reminded in 2.7.A. In short, a poroelastic material is entirely described by the porosity ϕ , the shear modulus of the frame N_s , the Poisson ratio ν , the density of the frame ρ_1 , the viscous and thermal characteristic lengths Λ and Λ' , tortuosity α_∞ , and the flow resistivity R_f when saturated by a light fluid^{94,95,98}. All these parameters are real valued except for the shear modulus, which incorporates a viscoelastic damping factor.

Introducing $\mathbf{s}^{(1)} = (\tilde{u}_1^s, \tilde{u}_2^s, \tilde{p})^\top$, the spatial Laplace transform of Eq. (2.13) is expanded in the form of Eq. (2.2) with

$$\begin{aligned} \mathcal{L}_0^{(1)} &= \begin{pmatrix} N_s \partial_{22} + \hat{\rho} \omega^2 & 0 & 0 \\ 0 & \hat{P} \partial_{22} + \hat{\rho} \omega^2 & \beta \partial_2 \\ 0 & -\beta / \phi^2 \partial_2 & 1/R + \partial_{22} / (\rho_{22} \omega^2) \end{pmatrix}, \\ \mathcal{L}_1^{(1)} &= \begin{pmatrix} 0 & i \hat{P} \partial_2 & i \beta \\ i(\hat{A} + N_s) \partial_2 & 0 & 0 \\ -i \beta / \phi^2 & 0 & 0 \end{pmatrix}, \text{ and } \mathcal{L}_2^{(1)} = \begin{pmatrix} \hat{P} & 0 & 0 \\ 0 & N_s & 0 \\ 0 & 0 & 1 / (\rho_{22} \omega^2) \end{pmatrix}. \end{aligned} \quad (2.15)$$

Finally, the discretized equations of motion read as

$$\underline{\underline{L}}^{(1)} \underline{\underline{s}}^{(1)} = \begin{pmatrix} (\omega^2 \hat{\rho} - k_1^2 \hat{P}) \underline{\underline{I}} + N_s \underline{\underline{D}}_{22} & i k_1 (\hat{P} - N_s) \underline{\underline{D}}_{22} & -i k_1 \beta \underline{\underline{I}} \\ i k_1 (\hat{P} - N_s) \underline{\underline{D}}_{22} & \hat{P} \underline{\underline{D}}_{22} + (\omega^2 \hat{\rho} - k_1^2 N_s) \underline{\underline{I}} & \beta \underline{\underline{D}}_{22} \\ -i \beta \omega^2 \underline{\underline{I}} & -\beta \omega^2 \underline{\underline{D}}_{22} & \frac{k_1^2}{\rho_{22}} \underline{\underline{I}} + \frac{\omega^2}{R} \underline{\underline{I}} \end{pmatrix} \underline{\underline{s}}^{(1)} = 0. \quad (2.16)$$

The equation of motion in the elastic medium is derived from the fundamental elasticity equations¹⁵⁴ and reads as,

$$\rho \omega^2 \mathbf{u}^{(2)} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}^{(2)}) + \mu \nabla^2 \mathbf{u}^{(2)} = 0, \quad (2.17)$$

where ρ is the density and λ and μ are the Lamé coefficients. The corresponding discretized equation of motion is

$$\underline{\underline{L}}^{(2)} \underline{\underline{s}}^{(2)} = \left(\frac{\rho^{(2)}}{\mu} \omega^2 \underline{\underline{I}} + \begin{pmatrix} -k_1^2(\lambda/\mu + 2) \underline{\underline{I}} + \underline{\underline{D}}_{22} & ik_1(\lambda/\mu + 1) \underline{\underline{D}}_{22} \\ ik_1(\lambda/\mu + 1) \underline{\underline{D}}_{22} & (\lambda/\mu + 2) \underline{\underline{D}}_{22} - k_1^2 \underline{\underline{I}} \end{pmatrix} \right) \underline{\underline{s}}^{(2)} = 0. \quad (2.18)$$

In addition, the same procedure can be followed for the equation of motion of a wave propagating in a layer of fluid material. We give its expression in order to provide an exhaustive list of the types of layers that can be modeled using the present method. The Helmholtz equation, $(\nabla^2 + k_0^2)p = 0$, with k_0 the bulk wavenumber and p the pressure field in the fluid material, is discretized into the form,

$$\left(\underline{\underline{D}}_{22} - k_1^2 + k_0^2 \right) \underline{\underline{p}} = 0. \quad (2.19)$$

The interface and boundary conditions depend on the vectors $\underline{\underline{I}}^\pm$, $\underline{\underline{D}}_2^\pm$ and $\underline{\underline{D}}_{22}^\pm$ which corresponds to the last (subscript +) and first row (subscript -) of the DMs. From the bottom to the top of the two-layer system, they are

— Rigid boundary condition at $\Gamma^{(0)}$ which leads to

$$\underline{\underline{u}}^{(1)}(x_2 = 0) = 0; \quad \underline{\underline{u}}_f^{(1)}(x_2 = 0) \cdot \underline{\underline{n}} - \underline{\underline{u}}_s^{(1)}(x_2 = 0) \cdot \underline{\underline{n}} = 0, \quad (2.20)$$

the discretized form of which is,

$$\underline{\underline{C}}^{(0)} \underline{\underline{s}}^{(1)} = \begin{pmatrix} \frac{0}{\underline{\underline{D}}_{22}} & \frac{\underline{\underline{I}}^+}{\underline{\underline{D}}_{22}} \\ \frac{\phi}{\rho_{22}} \underline{\underline{D}}_2^+ & \underline{\underline{D}}_{22}^+ - \left(1 + \frac{\rho_{12}}{\rho_{22}} \right) \omega^2 \underline{\underline{I}}^+ \end{pmatrix} \begin{pmatrix} \underline{\underline{p}}^{(1)} \\ \underline{\underline{u}}_1^{(1)} \\ \underline{\underline{u}}_2^{(1)} \end{pmatrix} = \underline{\underline{0}}. \quad (2.21)$$

— Continuity between the poroelastic and the elastic media¹⁵⁵ at $\Gamma^{(1)}$ which leads to

$$\underline{\underline{\sigma}}^{t(1)}(x_2 = b^{(1)}) - \underline{\underline{\sigma}}^{(2)}(x_2 = b^{(1)}) = 0; \quad \underline{\underline{u}}^{(2)}(x_2 = b^{(1)}) = \underline{\underline{u}}_s^{(1)}(x_2 = b^{(1)}); \quad (2.22)$$

$$\underline{\underline{u}}_f^{(1)}(x_2 = b^{(1)}) \cdot \underline{\underline{n}} - \underline{\underline{u}}_s^{(1)}(x_2 = b^{(1)}) \cdot \underline{\underline{n}} = 0,$$

where $\underline{\underline{\sigma}}^{t(1)} = \underline{\underline{\hat{\sigma}}}_s - \phi(1 + Q/R)$ is the total stress tensor in the poroelastic layer. The corresponding discretized interface matrices, arising from

$$\underline{\underline{C}}^{(1)+} \underline{\underline{s}}^{(2)} - \underline{\underline{C}}^{(1)-} \underline{\underline{s}}^{(1)} = \underline{\underline{C}}^{(1)} \begin{pmatrix} \underline{\underline{p}}^{(1)} & \underline{\underline{u}}_1^{(1)} & \underline{\underline{u}}_2^{(1)} & \underline{\underline{u}}_1^{(2)} & \underline{\underline{u}}_2^{(2)} \end{pmatrix}^\top = 0, \quad (2.23)$$

are cast as

$$\underline{\underline{C}}^{(1)} = \begin{pmatrix} -N_s \underline{\underline{D}}_2^+ & -ik_1 N_s \underline{\underline{I}} & \underline{\underline{0}} & \mu \underline{\underline{D}}_2^- & ik_1 \mu \underline{\underline{I}}^- \\ -ik_1 \hat{A} \underline{\underline{I}}^+ & -\hat{P} \underline{\underline{D}}_2^+ & \phi \left(1 + \frac{Q}{R} \right) \underline{\underline{I}}^+ & ik_1 \lambda \underline{\underline{I}}^- & (\lambda + 2\mu) \underline{\underline{D}}_2^- \\ \underline{\underline{0}} & -\underline{\underline{I}}^+ & \underline{\underline{0}} & \underline{\underline{I}}^- & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{0}} & -\underline{\underline{I}}^+ & \underline{\underline{0}} & \underline{\underline{I}}^- \\ -\frac{\phi}{\rho_{22}} \underline{\underline{D}}_2^+ & \left(1 + \frac{\rho_{12}}{\rho_{22}} \right) \omega^2 \underline{\underline{I}}^+ & \underline{\underline{0}} & \underline{\underline{0}} & \underline{\underline{0}} \end{pmatrix}, \quad (2.24)$$

with the left-hand side corresponding to $\underline{\underline{C}}^{(1)-}$ and the right-hand side to $\underline{\underline{C}}^{(1)+}$.

2.5. Application to a two-layer poroelastic-elastic structure

— Continuity between the elastic and the fluid media at $\Gamma^{(2)}$ which leads to

$$\begin{aligned}\sigma_{12}^{(2)}(x_2 = b) &= 0; & \sigma_{22}^{(2)}(x_2 = b) &= -p^{(0+)}; \\ u_2^{(2)}(x_2 = b) &= u_2^{(0+)}(x_2 = b).\end{aligned}\quad (2.25)$$

The corresponding discretized interface matrix $\underline{\underline{C}}^{(2)} \begin{pmatrix} \underline{u}_1^{(2)} & \underline{u}_2^{(2)} & A^{(0+)} \end{pmatrix}^T = 0$ is

$$\underline{\underline{C}}^{(2)} = \left(\begin{array}{cc|c} \underline{D}_2^+ & ik_1 \underline{I}^+ & 0 \\ ik_1 \lambda \underline{I}^+ & (\lambda + 2\mu) \underline{D}_2^+ & 1 \\ \hline 0 & \omega^2 \underline{I}^+ & \frac{ik_2^{(0)}}{\rho_f} \end{array} \right). \quad (2.26)$$

Combining the discretized interface conditions, Eqs.(2.21), (2.24) and, (2.26), with the discretized equations of motion, Eqs.(2.16) and (2.18), leads to the following system

$$\begin{pmatrix} \underline{\underline{C}}^{(0)} & 0 & 0 \\ \underline{\underline{L}}^{(1)} & 0 & 0 \\ \underline{\underline{C}}^{(1)-} & -\underline{\underline{C}}^{(1)+} & 0 \\ 0 & \underline{\underline{L}}^{(2)} & 0 \\ 0 & \underline{\underline{C}}^{(2)-} & -\underline{\underline{C}}^{(2)+} \end{pmatrix} \begin{pmatrix} \underline{s}^{(1)} \\ \underline{s}^{(2)} \\ A^{(0+)} \end{pmatrix} = \underline{0}. \quad (2.27)$$

This system is further expanded in the form of Eq.(2.7) and solved for a set of eigenvalues γ and associated eigenvectors $\mathcal{S}$ at each frequency ω . Wavenumbers k_1 are then obtained from γ . The dispersion curve is thus calculated over a given frequency range. However, the condition $\text{Re}(k_2^0) \geq 0$ should be imposed to sort out the solution that do not meet it.

Although usual and simple, this two-layer structure is challenging from a numerical point of view because it involves materials with very large impedance contrasts and layers whose thicknesses differ by several orders of magnitude. These problems are partially solved by normalizing each line and rearranging the terms of the matrix Eq. (2.7), leading to a better numerical conditioning of the system.

2.5.2. Dispersion relation

The material properties of the $b^{(1)} = 52$ mm-thick poroelastic (melanime) layer are listed in Table 2.1, while the properties of the $b^{(2)} = 1$ mm-thick aluminium plate are the density $\rho = 2700 \text{ kg.m}^{-3}$ and the Lamé coefficients $\lambda = 60.75 \text{ GPa}$ and $\mu = 26.03 \text{ GPa}$. The saturating fluid and that occupying the half-space is air with density $\rho_f = 1.213 \text{ kg.m}^{-3}$, the heat capacity ratio $\kappa = 1.4$, kinematic compressibility $\mu_f = 1.839 \times 10^{-5} \text{ Pa.s}$, the Prandtl number $\text{Pr} = 0.71$, and adiabatic bulk modulus κP_0 , where the atmospheric pressure is $P_0 = 1.013 \times 10^5 \text{ Pa}$. The total two-layer thickness is thus $b = b^{(1)} + b^{(2)} = 53 \text{ mm}$. $\mathcal{M}^{(1)} = 5$ and $\mathcal{M}^{(2)} = 11$ collocation points are employed respectively in the poroelastic and elastic layers.

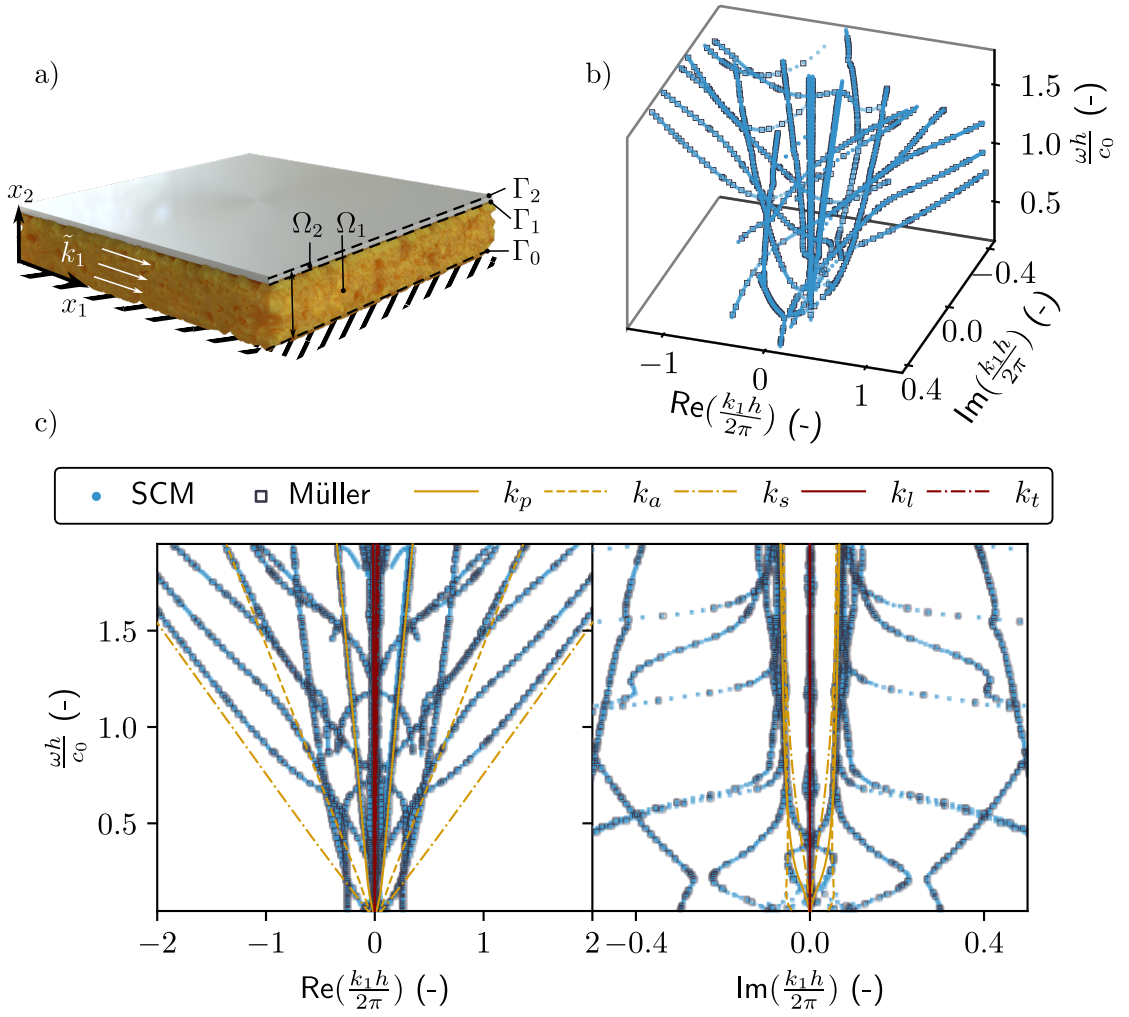


Fig. 2.2: a) Sketch of the two-layer structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. Blue dots represents the results as calculated with the SCM and black open markers represents those as calculated with the Müller method. Bulk wavenumbers in the poroelastic and elastic materials are highlighted in yellow and dark red respectively.

The 3D view of the complex dispersion diagram is depicted Fig. 2.2b. To ease readability, real and imaginary parts are depicted separately in Fig. 2.2c. Please note that this last representation can be misleading because only the modes that lie in the complex wavenumber range depicted in Fig. 2.2b are represented in Fig. 2.2c. Cut-on frequencies are thus fictitious. The results calculated using the SCM method, plotted in blue markers, are compared with those obtained using the Müller method, plotted with black open markers. Both methods provide identical results, up to around four digits, therefore validating the present approach. Note that the SCM solutions are used as initial guesses of the Müller algorithm to shorten computational time. In this way, fewer iterations are needed for the results of the root-finding method to converge. The main advantages of SCM are that no initial guesses are needed, i.e. as long as the discretization is sufficient to model the full wave behavior in the structure, the dispersion relation is guaranteed to be calculated. In addition, the matrices used for the numerical model are very small compared to those required by other methods such as FEM, and are therefore much less demanding in terms of calculation.

Wavenumbers are perfectly symmetric with both $\text{Re}(k_1 h / 2\pi) = 0$, and $\text{Im}(k_1 h / 2\pi) = 0$

2.5. Application to a two-layer poroelastic-elastic structure

Table 2.1: Parameters of the poroelastic material (melamine foam)

Parameter	ϕ	ρ_1	R_f	Λ	Λ'	α_∞	ν	N_s
Unit	-	kg.m ⁻³	kPa.s.m ⁻²	μm	μm	-	-	kPa
Melamine	0.98	6.5	5.6	214	214	1	0.24	11.96 (1 + i0.07)

axis, which is a feature of reciprocal systems. Wavenumbers having a very large slope and a low imaginary part correspond to modes mostly propagating in the aluminium plate. They are very close to the bulk longitudinal k_l and transverse k_t wavenumbers in the aluminum. This is due to the very large contrast between the elastic properties of the aluminum and those of the poroelastic frame. The other branches asymptotically tend toward the bulk acoustic, k_a , compression, k_p , or shear, k_s , wavenumbers of the poroelastic medium at high frequency. At low frequencies, these modes are highly dispersive.

If we compare these results with the dispersion relation depicted in 2.7.Bc) for the guided waves in a single poroelastic plate twice as thick and in the absence of the elastic plate, we find some essential differences. The branches associated with the elastic plate and the branch clearly associated with the coupling between the elastic and poroelastic layers (referenced as (1) in Fig.2.7.Bc)) do not exist. The other branches are similar to those of this single poroelastic layer, although slightly modified by the presence of the elastic plate. This proves a weak coupling between the elastic and poroelastic layers, particularly at high frequencies. At lower frequencies, the coupling between the two layers is stronger. Effectively, the wavelength is much larger than the thickness of at least one of the two layers in these frequency ranges, which favors coupling. The A_0 branch observed in the case of a single poroelastic layer disappear in the two-layer system rigidly backed, because this rigid-backing does not allow the existence of such a mode.

2.5.3. Mode shapes and energy flow

Beyond the SCM numerical efficiency, a significant advantage of the method is that the eigenvectors associated to each eigenvalue directly provides the associated mode shape. Since these mode shapes, i.e. eigenvectors, are calculated from an eigenvalue problem, their amplitude are defined up to a constant. They are normalized to the value of the pressure field at the bottom of the poroelastic layer, i.e. at the location of the rigid backing, because this condition imposes $\tilde{u}^{(1)}(0) = 0$ (and in particular $\tilde{u}_2^{(1)}(0) = 0$), which is equivalent to a maximum pressure field. From these normalized eigenvectors, any physical fields can be calculated as detailed in 2.7.C.

Let us consider two solutions $k_1^\pm = \pm |\text{Re}(k_1)|$ symmetric in the dispersion diagram with respect to the axis $\text{Re}(k_1) = 0$ as shown in Figs. 2.3a-b. The displacement fields and total energy fluxes in each direction are depicted in Fig. 2.3c-f. The results calculated using the SCM method, plotted in continuous lines, match those obtained using the Müller method, plotted with marker lines. These two modes are identical but propagate in opposite directions, i.e., the mode k_1^+ propagates towards positive x_1 while the mode k_1^- propagates towards negative x_1 . Displacement fields and energy fluxes are identical along the x_2 axis but are of equal modulus but opposite sign along the x_1 axis (see Fig. 2.3c-d and e-f). Both displacement field and energy flux amplitudes are two to three orders of magnitude smaller in the elastic plate (for $b_1 < x_2 < b$) than in the poroelastic plate. The aluminum coating is almost acting as a rigid

boundary condition, as also testified by the fact that the x_1 components of the displacement field and energy fluxes are almost symmetric with respect to the axis $x_2 = b_1/2$, while their x_2 components are almost antisymmetric along the x_2 axis. Note that the fluid phase Poynting vectors depicted in Fig. 2.3f are of equal modulus but opposite sign, symmetric with respect to the axis $x_2 = b_1/2$, and more than a hundred times smaller than total energy flux in amplitude. The energy of these modes are thus mainly localized in the solid phase of the poroelastic plate.

To go a step further, we evaluated the energy transport velocity $\bar{V}_e = \bar{P}_t / \bar{U}$ as the ratio between the average energy flux in each layer $\bar{P}_t$ over the mean total energy $\bar{U}$. The energy transport velocity of the mode highlighted in orange in Fig. 2.3a-b is compared to the group velocity $v_g = \partial\omega / \partial \text{Re}(k_1)$ ⁸⁶ in Fig. 2.3g. It is numerically computed by doing finite differences on a branch of the dispersion relation. Both velocities match for frequency ranges where the imaginary part of k_1 is low, i.e., for weak attenuation modes^{156–159}, but are different for strong attenuation. In particular, the group velocity diverges around the cut-on frequency. This is the reason why the energy transport velocity is usually preferred to study absorbing structures. The energy transport velocity $\bar{V}_e$ of the mode highlighted in green Fig. 2.3a-b is depicted in Fig. 2.3h. This energy transport velocity is always positive, while k_1 crosses the $\text{Re}(k_1) = 0$ axis. When $\text{Re}(k_1) < 0$ the phase velocity is negative, while the $\bar{V}_e > 0$. This branch (and the symmetric one) has also a negative group velocity as also testified by the fact $\text{Re}(k_1) < 0$ while $\text{Im}(k_1) > 0$ in the shaded area in Fig. 2.3h. This branch has also an exponentially growing amplitude.

2.5.4. Experimental validation

The experimental set-up is depicted in Fig. 2.4a. The sample is the two-layer structure considered in the previous subsection. The aluminum plate is glued on top of the poroelastic layer. The latter layer is glued on a 3 cm-thick aluminium plate, which mimic the rigid backing. The three elements are 45 cm wide and 85 cm long. The surrounding medium and the saturating fluid is air, at regular pressure and temperature conditions. The sample is positioned perpendicularly to the ground on its longest side as depicted in Fig. 2.4a to ease the mode excitation in the poroelastic layer. The excitation is provided by a shaker (Brüel and Kjaer type 4810), which is rigidly attached to the sample with a threaded steel rod fixed to the shaker on one side and glued on a 1mm thick aluminum plate of width 10 cm and height 1.5 cm. This plate is cut at the edge opposite to the threaded steel rod and glued to the porous sample, creating a line source at the edge of the sample. The resonance of this part was measured to be 4500 Hz, which is the upper limit of the measured frequency range. The excitation signal is a swept-sine function with 400 points ranging from 1 kHz to 3 kHz. The general layout of the experiments is similar to that used originally in¹⁶⁰. The key difference in our case is that the excitation is operated directly to the poroelastic layer and not to the top of the aluminum plate, because the modes that corresponds to propagating solutions in the poroelastic layer are encapsulated in between two almost rigid plates (see Section 2.5.3). The normal displacement is measured with a laser Doppler vibrometer (Polytec VibroFlex Neo) on a line of length $L = 40$ cm, along the side of the thickness of the porous layer. These measurements are averaged 100 times. The excitation signal and the measured field are interfaced through a computer via a Zürich Instruments acquisition card.

The SLaTCoW method is used to analyze the space-frequency displacement measurements and recover the complex wavenumber-real frequency dispersion relation. This method relies

2.5. Application to a two-layer poroelastic-elastic structure

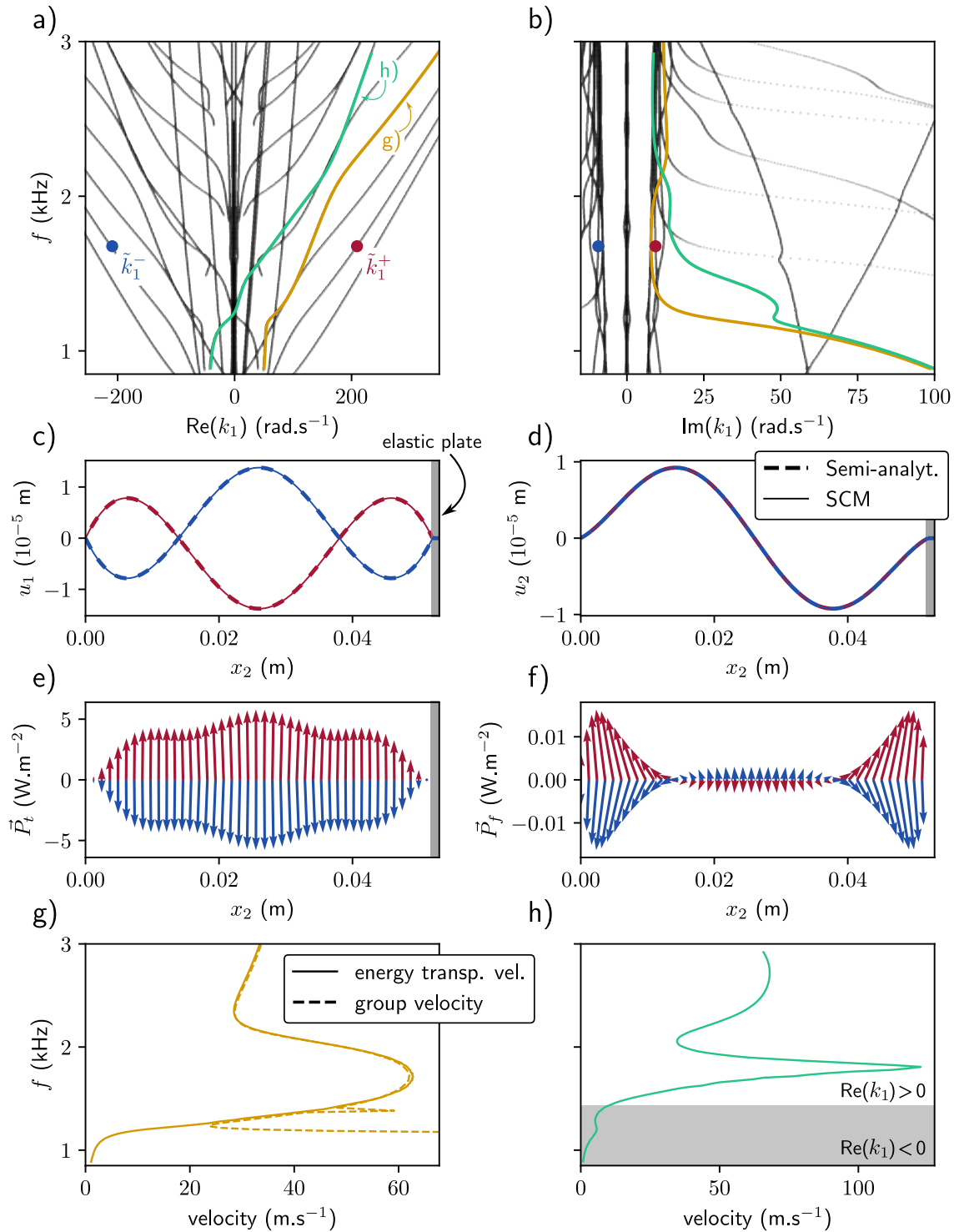


Fig. 2.3: a) and b) represent the real and imaginary parts of the dispersion relation of the two-layer structure.

Two branches are highlighted in yellow and green, as well as two specific solutions, $\tilde{k}_1^-$ (blue) and $\tilde{k}_1^+$ (red), symmetric with respect to the $\text{Re}(k_1) = 0$ axis. Displacement fields in the x_1 c) and x_2 direction d) associated with the $\tilde{k}_1^+$ and $\tilde{k}_1^-$ solutions displayed. Total Poynting e) vector and fluid Poynting vector f) with the length of the arrow indicating the amplitude and angle indicating the direction of the vector field. g) Group and energy transport velocities for the yellow branch. h) Energy transport velocity for the green branch. The gray area represents the frequency range over which this velocity is negative.

on the Laplace spatial transform of these measurements, which gives access to the real and imaginary components of the complex wavenumber. The difference between this Laplace transform and a correctly chosen ansatz function is finally minimized for each frequency, to recover the dispersion relation. More details on this method are given in Appendix 2.7.E as well as in the original article¹⁶⁰. The complex wavenumber-real frequency dispersion relation in Fig. 2.4b-c depicts the wavenumbers recovered experimentally k_s and simulated k_n . Four modes were sought during the minimization with the SLaTCow method: three of them correspond to the three bulk waves due to excitation and the fourth is depicted. The experimentally recovered mode jumps from one branch to the other because of the large modal density. The cut-on frequencies of the modes are clearly visible, especially on the imaginary part of the wavenumber. The lower real wavenumber parts of the numerical dispersion relation are not retrieved with the experimental results. This is because they correspond to a high attenuation of the mode, a part that is inherently difficult to measure in this experimental conditions. Generally, the recovered branches are those around the bulk solid compression wavenumber, corresponding to the wave that we most probably excite in the structure. The difficulties in efficiently exciting all the guided waves supported by this configuration largely explain the missing curves in the dispersion relation. The location and orientation of the excitation line limit possible symmetries and the generation of potential shear waves. This choice was made to avoid the main excitation of guided waves in the elastic plate when excitation and displacement measurement are performed only on this element. In addition, the modes mainly supported by the fluid phase should also be better excited by a loudspeaker and therefore better measured by a microphone. However, we have chosen to focus on the modes mainly supported by the solid phase. Nevertheless, the agreement between experimental results and simulation is quite good, thus validating the method.

2.6 | Conclusion

A spectral collocation method (SCM) is proposed to calculate dispersion relations for guided acoustic wave propagation in multilayer structures. The formalism is general enough to be able to consider arbitrary arrangements of (visco-) elastic, poroelastic, and (viscothermal-) fluid layers. Outgoing fluid radiation, leading to Leaky modes, can be accounted for. The fields in each layer is approximated by Chebyshev polynomials and discretized on the corresponding nodes. Interface conditions are implemented to couple layers together. The solution provides the full complex wavenumber-real frequency dispersion relation of the considered structure. In addition, the physical fields in each layer are directly evaluated since they are the eigenvectors associated with the eigenvalue problem giving the dispersion relation. The dispersion diagram calculated with the present SCM is validated against experimental results and that calculated with a root-finding method (Müller method) in a simple two-layer structure consisting of a rigidly backed poroelastic layer covered by a thin aluminum plate radiating in a fluid half-space. The semi-analytical approach underlying the root-finding method is taken a step further to solve for the wave amplitudes as well, providing a reference for the mode shapes calculated with the SCM. The dispersion properties in such structures, as well as the direction of the energy density in each phase of the poroelastic layers using a decoupled expression for the Poynting vectors is analyzed. Results on energy transport velocity are also reported.

This method paves the way of complex wavenumber-real frequency dispersion relation cal-

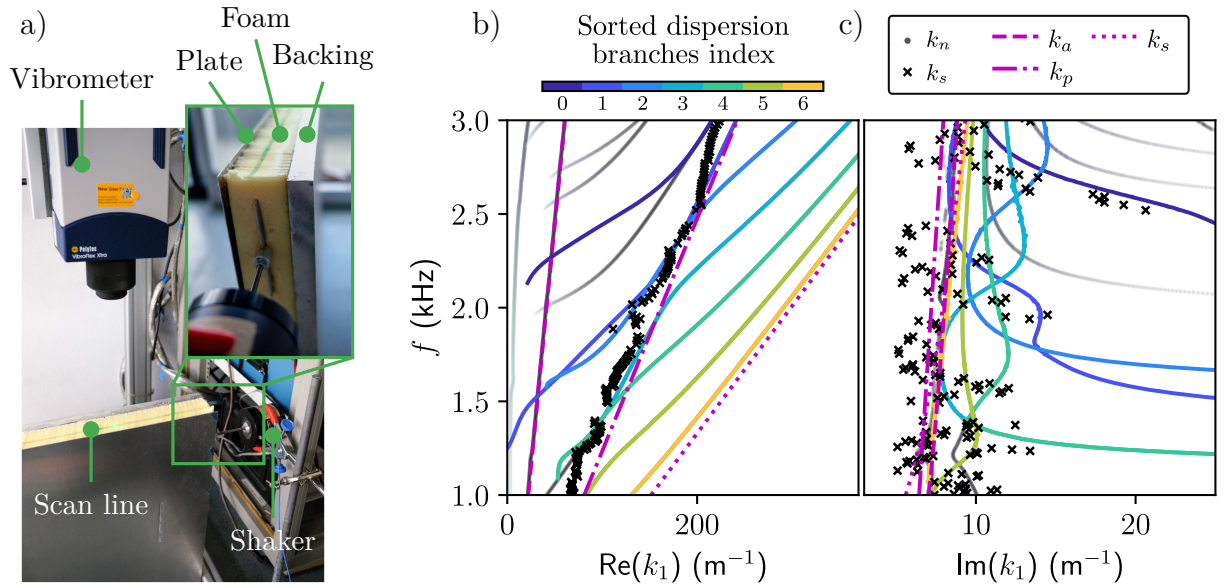


Fig. 2.4: a) Experimental set-up used for the measurement of the normal displacement field of the two-layer structure, with a close-up of the sample excitation. b) Real and c) imaginary parts of the dispersion relations recovered experimentally from the SLaTCoW method k_s (black crosses) and calculated from the SCM k_s (colored lines and black-to-grey dots) and the bulk wavenumbers in the poroelastic material (magenta lines).

ulation and analysis of more complex structures, like layers with embedded periodic inclusions (elastic or resonant), i.e. metaporoelastic surfaces.

2.7 | Appendices

2.7.A. Expression of the Biot elastic coefficients and effective parameters in poroelastic materials

The complex and frequency dependent density and bulk modulus of the fluid phase, which accounts respectively for the viscous and thermal losses are^{94,95} as,

$$\begin{aligned} \rho_{eq}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi} \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \omega} \sqrt{1 - i \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda} \right)^2} \right), \\ K_{eq} &= \frac{\kappa P_0}{\phi} \left(\kappa - (\kappa - 1) / \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \text{Pr} \omega} \sqrt{1 - i \text{Pr} \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right) \right)^{-1}, \end{aligned} \quad (2.28)$$

where ϕ is the porosity, α_∞ is the tortuosity, Λ and Λ' are respectively the viscous and thermal characteristic lengths, R_f is the flow resistivity, μ_f is the dynamic viscosity, κ is the heat capacity ratio, ρ_f the density of the saturating fluid, and P_0 is the ambient pressure.

The Biot elastic coefficients, P , Q , and R , are more commonly expressed in terms of the frame, K_b , effective fluid, K_{eq} , and shear, N_s , moduli when saturated by a light fluid as

$$P = K_b + \frac{4}{3}N_s + (1 - \phi)^2 \frac{K_{eq}}{\phi}; \quad Q = K_f(1 - \phi), \quad R = \phi K_{eq}, \quad (2.29)$$

while the apparent densities are

$$\rho_{22} = \phi^2 \rho_{eq}, \quad \rho_{12} = \phi \rho_f - \rho_{22}, \quad \rho_{11} = (1 - \phi) \rho_s - \rho_{12}, \quad (2.30)$$

with ρ_s the density of the solid phase. The coefficients in Eq. (2.13) are thus

$$\hat{A} = K_b - \frac{2}{3}N_s + \frac{K_{eq}}{\phi}((1 + (1 - \phi))), \quad \hat{\rho} = \rho_1 + \phi \rho_f - 2\phi^2 \rho_{eq} - \frac{\rho_f^2}{\rho_{eq}} \quad \beta = \frac{\rho_f}{\rho_{eq}} - 2\phi - 1, \quad (2.31)$$

where $\rho_1 = \phi \rho_f + (1 - \phi) \rho_s$ is the apparent density of the frame. Finally, the frame bulk modulus and shear modulus are linked by the Poisson ratio ν with $K_b = 2N_s(1 + \nu)/3(1 - 2\nu)$.

2.7.B. Application to a single poroelastic layer surrounded by two fluid half-spaces

The dispersion relation of guided waves in a single layer of poroelastic material surrounded by two identical fluid half-spaces is considered (See Fig. 2.5a). The poroelastic material properties are those considered in Section 2.5.2 and the layer thickness is $2b^{(1)} = 104$ mm, i.e. twice the thickness of the poroelastic layer considered in the two-layer system.

The discretized equations of motion is that provided in Eq. (2.16). $M^{(1)} = 11$ collocation points are considered. Only the boundary conditions at the lower, i.e. Γ_0 at $x_2 = 0$, and upper, i.e. Γ_1 at $x_2 = b$, interfaces are modified. They read as¹⁵⁵

$$\begin{aligned} \left(1 + \phi + \phi \frac{\rho_{12}}{\rho_{22}}\right) u_2^{(1)} + \frac{\phi^2}{\rho_{22} \omega^2} \partial_2 p^{(1)} &= u_2^{(0\pm)}, \quad p^{(1)} = p^{(0\pm)}, \\ \hat{\sigma}_{22}^s &= \left(1 - \phi \left(1 + \frac{Q}{R}\right)\right) p^{(0\pm)}, \quad \hat{\sigma}_{12}^s = 0. \end{aligned} \quad (2.32)$$

The discretized form of these boundary conditions is

$$\underline{C}_1 = \begin{pmatrix} \frac{\phi^2}{\rho_{22}} \underline{D}_2^+ & \underline{0} & \omega^2 \left(1 + \phi \left(1 + \frac{\rho_{12}}{\rho_{22}}\right)\right) \underline{I}^+ & \frac{ik_2^{(f)}}{\rho_f} \\ \underline{I}^+ & \underline{0} & \underline{0} & -1 \\ \underline{0} & ik_1(\hat{P} - 2N_s) \underline{I}^+ & \hat{P} \underline{D}_2^+ & 1 - \phi \left(1 + \frac{Q}{R}\right) \\ \underline{0} & N_s \underline{D}_2^+ & ik_1 N_s \underline{I}^+ & 0 \end{pmatrix}. \quad (2.33)$$

These matrices are then arranged in the same manner as Eq. (2.10) and the wavenumbers are calculated. Fig. 2.5b-c depicts the results.

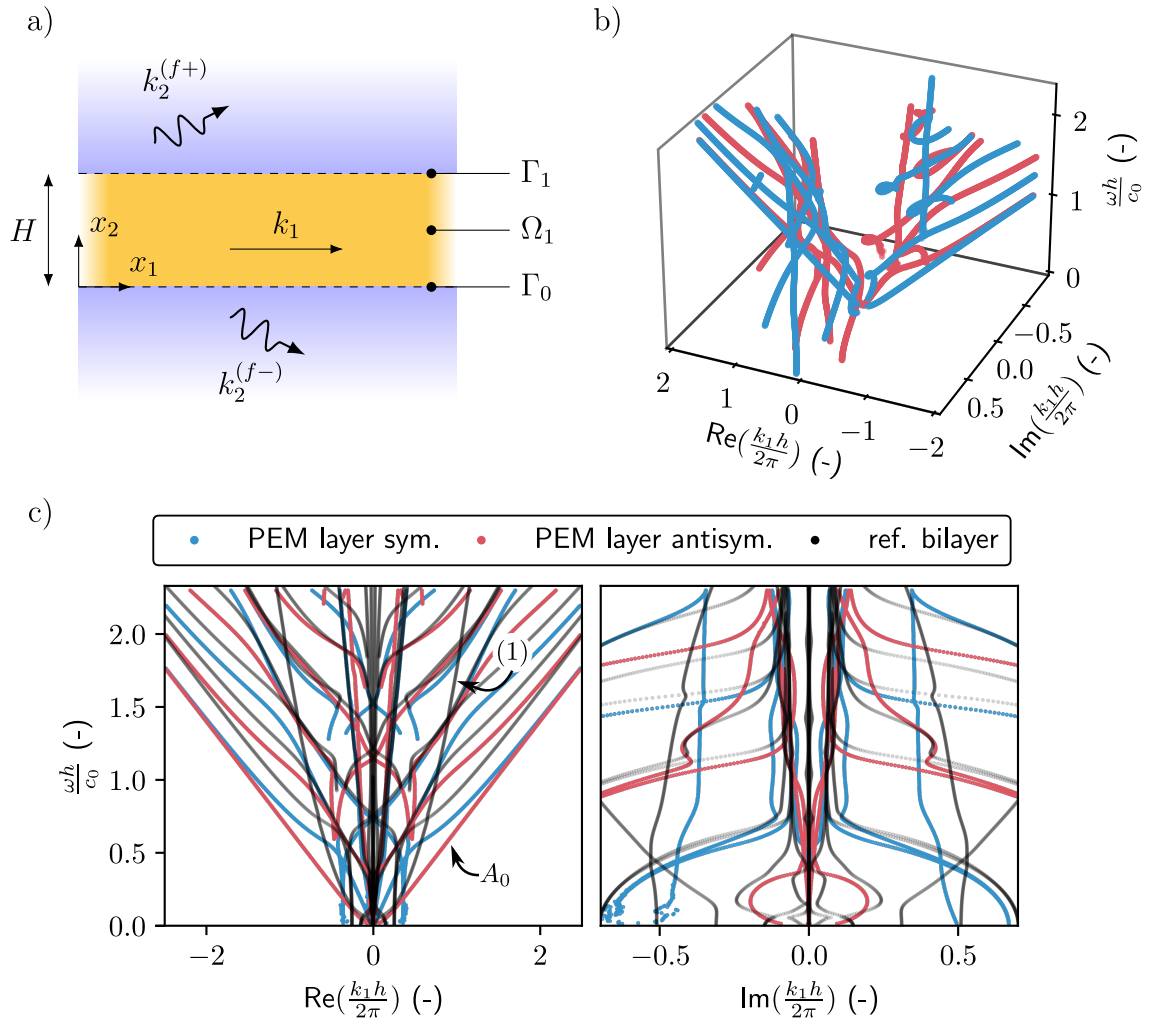


Fig. 2.5: a) Sketch of the single layer poroelastic structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. The branches of the symmetric and antisymmetric modes are represented by blue and red curves respectively. The dispersion relation of the two-layer elasic-poroelastic system studied in Section 2.5 are reminded with black lines.

In a similar way as Lamb modes, the different modes can be sorted in symmetric and antisymmetric modes. This is done by comparing the sign of the radiated amplitudes at each edge of the layer. Results are shown in Fig. 2.5b. All modes necessarily have an imaginary part, both because of the radiation condition, i.e. leakage, and because of viscothermal dissipation as can be noticed in Fig. 2.5c.

2.7.C. Evaluation of the physical fields from the SCM solutions

The discrete sets of eigenvectors resulting from the SCM are the spatial Laplace transforms of some physical fields at collocation points, i.e. the locations where the discrete equations of motion (residue) are exactly satisfied. These discrete fields are interpolated into a continuous form using

$$s^{(n)}(x_2) = \left[\psi_m^{-1}(\underline{\xi}) s_m^{(n)}(\underline{\xi}) \right] \psi(\xi^{(n)}) \quad (2.34)$$

where the terms in brackets correspond to the coefficients $\alpha_m^{(n)}$ given in Eq. (2.8). The Chebyshev polynomials ψ is the basis that maps from the discrete grid $\underline{\xi}$ back to the physical space.

The stress tensor components depend on the direct expressions of the fields and their first order derivative in x_2 , denoted with a prime symbol in the following. The various fields, differentiated up to any order, can be calculated by replacing ψ by its derivative in Eq. (2.34). Explicit expressions are given in the following,

- in the elastic layer, the spatial Laplace transforms of stress tensor components are computed from $\tilde{\mathbf{u}}$, with

$$\begin{aligned} \tilde{\sigma}_{11} &= (\lambda + 2\mu)ik_1\tilde{\mathbf{u}}_1 + \lambda\tilde{\mathbf{u}}'_2, & \tilde{\sigma}_{12} &= \mu(\tilde{\mathbf{u}}'_1 + ik_1\tilde{\mathbf{u}}_2), \\ \tilde{\sigma}_{22} &= \lambda ik_1\tilde{\mathbf{u}}_1 + (\lambda + 2\mu)\tilde{\mathbf{u}}'_2. \end{aligned} \quad (2.35)$$

- in the poroelastic layer, the full solid stress tensor is expressed as $\tilde{\sigma}_s = \hat{\sigma}_s - \phi(Q/R)\tilde{\mathbf{p}}$, with $\hat{\sigma}_s$ the spatial Laplace transform of the *in vacuo* solid stress tensor introduced in Section 2.5. The components of this full solid stress tensor are,

$$\begin{aligned} \tilde{\sigma}_{s,11} &= P ik_1 \tilde{\mathbf{u}}_{s,1} + \hat{A} \tilde{\mathbf{u}}'_{s,2} - \phi \frac{Q}{R} \tilde{\mathbf{p}}, & \tilde{\sigma}_{s,12} &= N_s (\tilde{\mathbf{u}}'_{s,1} + ik_1 \tilde{\mathbf{u}}_{s,2}), \\ \tilde{\sigma}_{s,22} &= \hat{A} ik_1 \tilde{\mathbf{u}}_{s,2} + P \tilde{\mathbf{u}}'_{s,2} - \phi \frac{Q}{R} \tilde{\mathbf{p}}. \end{aligned} \quad (2.36)$$

The Laplace transform of the fluid stress tensor is calculated as

$$\tilde{\sigma}_f = -\phi \tilde{\mathbf{p}} \delta_{ij}, \quad (2.37)$$

and the fluid displacement components are

$$\tilde{\mathbf{u}}_{f,1} = \frac{\phi}{\rho_{22}\omega^2} ik_1 \tilde{\mathbf{p}} - \frac{\rho_{12}}{\rho_{22}} \tilde{\mathbf{u}}_{s,1}, \quad \tilde{\mathbf{u}}_{f,2} = \frac{\phi}{\rho_{22}\omega^2} \tilde{\mathbf{p}}' - \frac{\rho_{12}}{\rho_{22}} \tilde{\mathbf{u}}_{s,2}. \quad (2.38)$$

All physical fields $\theta(\mathbf{x})$ can finally be calculated by the inverse spatial Laplace transform with $\theta(\mathbf{x}) = \tilde{\theta}(k_1, x_2) e^{ik_1 x_1}$.

2.7.D. Energy flux in the multilayer structure - Poynting theorem in a poroelastic medium

Energy conservation is commonly conducted in the time domain. Let us consider a volume Ω , bounded by $\partial\Omega$, with $\mathbf{n}$ the outgoing normal vector. The energy balance or Poynting theorem in a poroelastic materials reads as^{98,161}

$$\begin{aligned} - \int_{\partial\Omega} (-\boldsymbol{\sigma}_s \cdot \mathbf{v}_s^*) \cdot \mathbf{n} + (-\boldsymbol{\sigma}_f \cdot \mathbf{v}_f^*) \cdot \mathbf{n} dS &= \frac{1}{2} \int_{\Omega} \rho_1 \partial_t |\mathbf{v}_s|^2 - \rho'_{12} \partial_t (\mathbf{v}_f - \mathbf{v}_s)^2 + \rho_2 \partial_t |\mathbf{v}_f|^2 d\Omega \\ &+ \int_{\Omega} \boldsymbol{\sigma}_s : \partial_t \boldsymbol{\epsilon}_s^* + \boldsymbol{\sigma}_f : \partial_t \boldsymbol{\epsilon}_f^* - b (\mathbf{v}_s - \mathbf{v}_f)^2 d\Omega, \end{aligned} \quad (2.39)$$

where $\mathbf{v}_s = \partial_t \mathbf{u}_s$ and $\mathbf{v}_f = \partial_t \mathbf{u}_f$ are the velocity fields, $\boldsymbol{\sigma}_s$ and $\boldsymbol{\sigma}_f$ are the stress tensors, $\boldsymbol{\epsilon}_s = 1/2 (\nabla \mathbf{u}_s + \nabla^T \mathbf{u}_s)$ and $\boldsymbol{\epsilon}_f = 1/2 (\nabla \mathbf{u}_f + \nabla^T \mathbf{u}_f)$ are the strain tensors in the solid frame and in the fluid phase respectively, and $\star$ denotes the complex conjugate. Note that a similar expression is provided in¹⁵⁹, where the alternative 1962 Biot formulation is used.

The left-hand side contains the energy density flux $\mathbf{P}_s = -\boldsymbol{\sigma}_s \cdot \mathbf{v}_s^*$ and $\mathbf{P}_f = -\boldsymbol{\sigma}_f \cdot \mathbf{v}_f^*$ in the elastic frame and fluid phase. The right-hand side contains the kinetic energy K ¹¹ and the internal forces $P_{\text{int}} = \partial_t W + P_{\text{dis}}$. The latter is the sum of the time derivative of the strain energy W and the dissipated power P_{dis} by the elastic damping and the viscous and thermal losses¹⁶¹. In the frequency domain, the kinetic and strain energies read as

$$\begin{aligned} K &= \frac{1}{2} \rho_1 \partial_t |\mathbf{v}_s|^2 - \frac{1}{2} \rho'_{12} \partial_t (\mathbf{v}_f - \mathbf{v}_s)^2 + \frac{1}{2} \rho_2 \partial_t |\mathbf{v}_f|^2, \\ W &= \frac{1}{2} \hat{\boldsymbol{\sigma}}_s : \boldsymbol{\epsilon}_s^* + \frac{1}{2} R \nabla \cdot (Q \mathbf{u}_s + R \mathbf{u}_f), \end{aligned} \quad (2.40)$$

Finally, the time average total Poynting vector is $\langle \mathbf{P}_t \rangle = \langle \mathbf{P}_s \rangle + \langle \mathbf{P}_f \rangle$, with $\langle \mathbf{P}_s \rangle = -\text{Re} (\boldsymbol{\sigma}_s \cdot \mathbf{v}_s^*) / 2$ and $\langle \mathbf{P}_f \rangle = -\text{Re} (\boldsymbol{\sigma}_f \cdot \mathbf{v}_f^*) / 2$ and the total energy density is $\langle U \rangle = \langle K \rangle + \langle W \rangle = \text{Re}(K)/2 + \text{Re}(W)/2$. Note that a similar procedure can be followed for isotropic elastic or fluid medium¹¹⁴. In particular, the elastic Poynting vector is $\langle \mathbf{P} \rangle = \text{Re} (\boldsymbol{\sigma} \cdot \mathbf{v}^*) / 2$, where $\boldsymbol{\sigma}$ is the stress tensor and $\mathbf{v}$ is the velocity in the elastic medium.

The energy transport velocity V_e is evaluated from the latter expressions

$$V_e = \frac{\langle \mathbf{P}_t \rangle \cdot \mathbf{n}}{\langle U \rangle}. \quad (2.41)$$

When a multi-layer configuration is considered, the latter velocity is averaged over the structure thickness $\bar{V}_e = \frac{\bar{P}_t}{\bar{U}}$, with the average Poynting vector and the average total energy defined as

$$\bar{P}_t = \sum_n \frac{1}{h_n} \int_{h_{n-1}}^{h_n} \langle \mathbf{P}_t^{(n)} \rangle \cdot \mathbf{e}_{x_2} dx_2, \quad \bar{U} = \sum_n \frac{1}{h_n} \int_{h_{n-1}}^{h_n} \langle U^{(n)} \rangle dx_2. \quad (2.42)$$

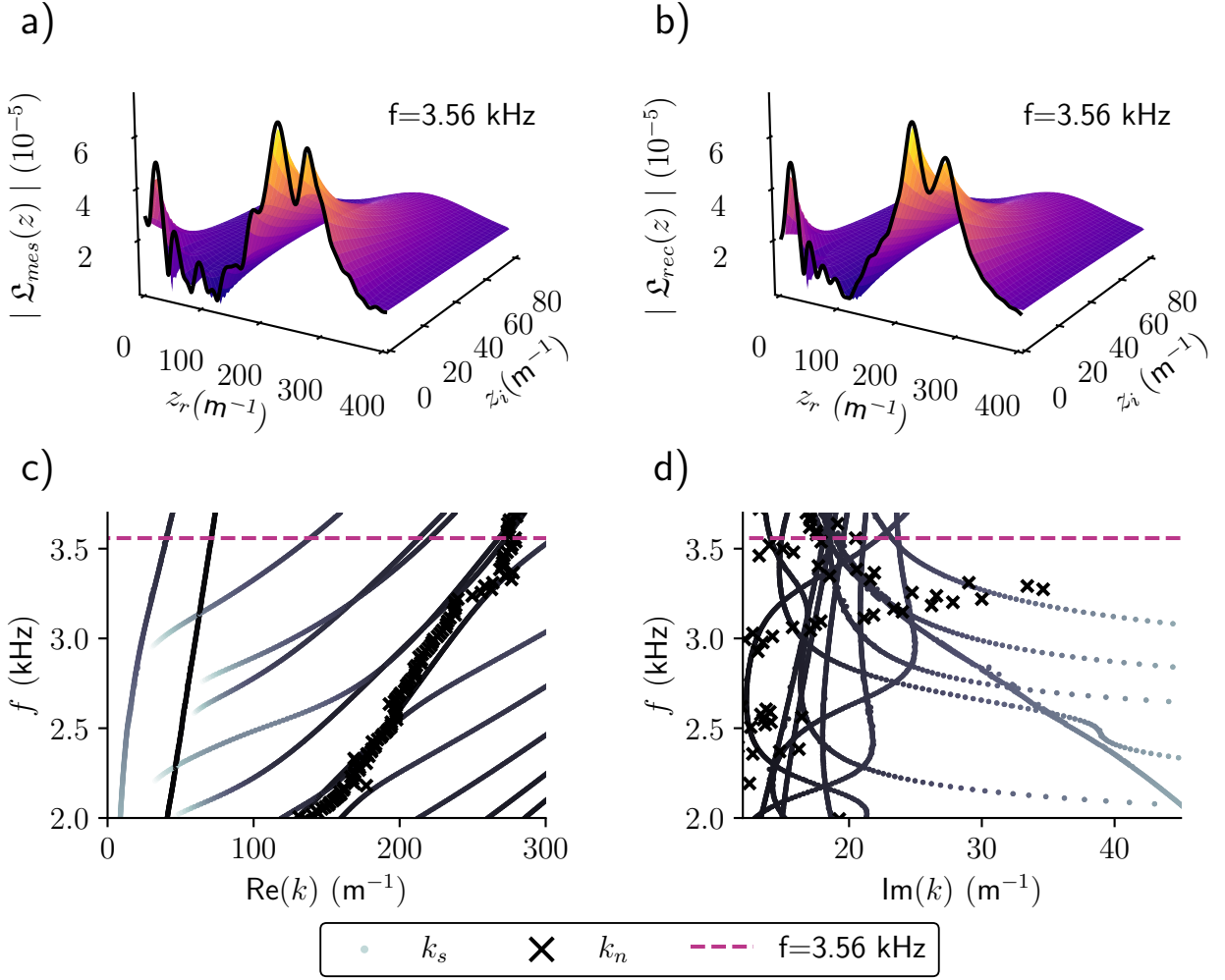


Fig. 2.6: Description of the procedure of the SLaTCoW method: a) Spatial Laplace transform from the experimental data, b) Spatial Laplace transform of the reconstructed wave fields. Real c) and imaginary d) parts of the experimentally retrieved wavenumbers (crosses) compared to the SCM (black-to-grey dots). The frequency at which the Laplace transforms are represented corresponds to the magenta dashed line in the graph.

2.7.E. Reminder of the SLaTCoW method for obtaining complex-valued wavenumbers.

The key point of the SLaTCoW method is the Laplace transform, $\mathfrak{L}_{\text{mes}}(z)$ with $z = z_r + iz_i$ the Laplace space coordinate, of the data measured in space at a fixed frequency ω . In doing so, we move from the spatial domain to the complex wavenumber domain. The difference between $\mathfrak{L}_{\text{mes}}(z)$ and the spatial Laplace transform of the displacement ansatz along the same line, $\mathfrak{L}_{\text{rec}}(z)$, is then minimized in the least square sense to recover both the complex amplitude (magnitude and phase) and the complex wavenumber of each mode assumed to be part of the displacement ansatz.

The experimental Laplace spatial transform $\mathfrak{L}_{\text{mes}}(z) = \int_0^L r(x)e^{-izx}dx$ is computed at each frequency ω from the transfer function $r(x)$ between the excitation and the normal dis-

placement measured along a line $x \in [0, L]$, with $L = 0.3$ m and spacing $d_x = 0.001$ m. The displacement anstaz is assumed to be of the form $w(x) = \sum_{\beta} A^{\beta} e^{ik^{\beta}x}$, with β the number of guided-waves, i.e., the number of modes, and where $A^{\beta} = |A^{\beta}|e^{i\phi^{\beta}}$ is the complex amplitude and k^{β} is the complex wavenumber of the β -th mode.. In practice, β is the smallest integer that gives a good fit between the two Laplace transforms and therefore depends on the frequency under consideration. Its spatial Laplace transform reads as $\mathfrak{L}_{\text{rec}}(z) = \sum_{\beta} A^{\beta} L e^{i(k^{\beta}-z)\frac{L}{2}} \text{sinc}\left((k^{\beta}-z)\frac{L}{2}\right)$.

The cost function $\mathfrak{C} = (\|\mathfrak{L}_{\text{rec}}(z) - \mathfrak{L}_{\text{mes}}(z)\|) / \|\mathfrak{L}_{\text{mes}}(z)\|$ is finally minimized using the Matlab function *fminsearchbnd* for $\{|A^{\beta}|, \phi^{\beta}, \text{Re}(k^{\beta}), \text{Im}(k^{\beta})\}, \forall \beta$. The initial guess are the recovered values of the minimization parameters at the previous frequency, with the exception of the first frequency. The method is graphically exposed in Figure 2.6. Figures 2.6 c) and d) depict the complex dispersion relation as already depicted in Fig. 2.4. Figure 2.6a) depicts the spatial Laplace transform of the experimental data and Figure 2.6b) depicts $\mathfrak{L}_{\text{rec}}(z)$ with the parameters derived from minimization of the cost function at the frequency $f = 3.56$ kHz. This frequency is highlighted by the purple dashed line in Figures 2.6 c) and d). The two Laplace transforms match almost perfectly.

Conclusion & Perspectives

Ranging from a stable method for the evaluation of the scattering properties of grating stacks to developing ways to compute leaky guided waves in homogeneous multilayer structures as well as in metaporous layers, this thesis work has thus been mainly focused on the development of numerical tools.

Throughout the literature, these metaporous structures have shown efficient absorption properties using various geometry by embedding resonant, tortuous paths, voids or inclusions inside porous and poroelastic layers. These obstacles provide additional dispersion as well as interferences of the wave at discrete frequencies. Combined with the attenuation of waves in porous layer, they provide for efficient acoustic absorption structures. Even though, these effects have been studied in the literature, they have mostly been considered as infinite systems. In this work, the considerations were extended to the study of leaky guided waves in these dissipative and highly-dispersive structure, that is to account for leakage from the structure to surrounding media. This has shown that the classification of the waves was of interest and that the physical mechanisms at stake were not so simple. The combination of these dissipation mechanisms thus have showed a complex behaviour for the studied structure.

In CHAPTER 1, we laid the foundation for the work, introduced the governing motion equations for assumptions of homogeneous porous and poroelastic media and proposed a state of art for the existing numerical method allowing for the computation of the scattering properties as well as the dispersion of waves in these structures.

From this, in CHAPTER 2 we discuss as a first part a simple dissipative homogeneous (per layer) multilayer structure with outgoing radiation. These homogeneous layers can easily be discretized in a 1D stencil, using the SCM. In a general formalism, any stack of layer, dissipative or not, can be computed with the presented method. Results regarding the energy flux for specific modes were presented as well

Then, we developed a robust and stable method for the main configurations of interest in this work, that are grating stacks with poroelastic/porous layers as presented in CHAPTER 3. The method remained stable for all tested cases contrary to the TMM and the coupling is thus quite robust and efficient, since no fine discretization of the homogeneous layers is required.

In the final CHAPTER 4 we adapted the previous method, in the absence of a source, solving its dispersion relation. The method relies on the same discretization as previously, therefore the root-finding solving of the method is not very efficient. However, reliable results have still been achieved. We were able to highlight the trapped mode that occurs in a perfect absorption case as well as the impact of the resonance due to an hybridization of the resonator and the cavity for split-ring resonators.

Perspectives and future work

1. The finite elements approach developed and presented in Chapter 3 has been tested and validated for general layers. However, in Chapter 4, dispersion relations are only studied for the case of equivalent fluid; future work thus include an extension of the study of dispersion relation of waves in metaporous structures, by assuming poroelastic layers and solid inclusions, leading to a much larger of modes to analyse due to the presence of the modes linked to the elastic phase of the materials modeled with the full Biot theory.

2. An extension to 3D structures can be thought of for all the studied configurations in this thesis. Allowing to solve also the out-of-plane modes could allow for the study of 2D phononic crystals with leakage in surrounding space and provide more insight on real-life applications. Also this could mean that the structures could be finite in an additional direction.

3. Regarding the Global Method that was derived and benchmarked against TMM, an extension with the Recursive Method, formalized in¹⁶² was also worked on. Unfortunately, it has ended up on a dead end, because even if the original authors had succeeded in a stabilized formulation for homogeneous stacks of layers, the coupling with finite elements revealed to still be unstable. The investigation of the stability of all these methods could be an interesting perspective of focus.

4. An important perspective for the metaporous structure dispersion relation computation presented in Chapter 4 would involve the development of a more robust method not relying on a root-finding procedure. Some time has been spent during the development of the method to try to write the equations in the form of an eigenvalue problem with k_{1q} as the eigenvalue, although this was not achieved due to the challenges of obtaining such a formulation with the way the coupling between the surrounding medium and the structure is done. The finite elements matrices obtained in Eq. [ref the correct eq](#) depend on both on a term in $e^{ik_{1q}d}$ and an integral on x_1 with a $e^{ik_{1q}x_1}$ term. An appropriate change of variable or a different formulation for the coupling might provide a way to achieve the writing of an eigenvalue problem. Note that this has been explained briefly in Section. [ref annexe PRB](#) and shows that the eigenvalue problem is easily written when not accounting for the surrounding medium.

5. A study the excitability of the modes could be conducted by using a numerical model done with COMSOL & a line source excitation properly tuned according to the dispersion relation modes to confirm the form of the solutions of the modes in finite structures and a more *realistic* configuration and further link the dispersion relation results to an actual observable behaviour of the structures by gaining insights on the excitability of the modes. This is an important aspect as there are many dissipation mechanisms involved, leading to very attenuated modes: seeing if they can be excited or if their existence actually affects the wave propagation in any way.

6. An experimental validation of these results would add value to this work and could complement the previous point. One could imagine a porous sample manufactured with rigid cylindrical inclusions. There would be challenges to this, including the precision required to engineer precisely the radius of the cylinders. Measurements need to take into account the nature of the source needed to excite the variety of these modes. A post-process of the frequency responses using SLaTCoW or some other method allowing for the retrieval of complex wavenumbers. This approach is similar to that applied in Section.??.

Appendices

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A | Implementing SAFE to compute the dispersion relation of Lamb modes

This section has initially been worked on as part of a letter to the reviewers of the paper^{P1} dealing with the SCM and added to Chapter 1. They wanted us to investigate the performance of SCM against that of SAFE, which we have done in a simple configuration, that of the free elastic plate. This work was carried out for the purpose of providing clarification for reviewers and was decided to not be included in the article. We provide it as the following appendix.

As discussed in Section. 2.3, the SAFE method is an alternative to SCM for the numerical computation of dispersion relation of guided waves in layers. They rely on a similar formalism, that is a 1D discretization across the thickness of each layer and an assumption of guided harmonic waves along the longitudinal direction. An analogous method to that presented in Chapter 2 could be developed using SAFE, the key difference being the location of nodes in the layer and the interpolation function used for discretization. There is a large bibliography dealing with these topics and applying this method to similar problem to ours.

The general theory of SAFE is explained in great details in Chapter 9 of Rose's textbook¹⁴⁰ hence the present method is inspired from. Examples of SAFE application to dispersion curve calculation are illustrated by Predoi *et al.*¹⁶³ where a numerical model was implemented to investigate periodic structure of infinite width with the addition of periodic boundary

conditions allowing for the computation of guided wave modes. A similar approach was generalized for systems with arbitrary cross-sections and a coupling with infinite solid media by using the so-called absorbing region¹⁶⁴. Other description of the surrounding medium where leakage happens have also been investigated by using a PML¹¹⁸. This has been done by Mazzotti *et al.* using BEM^{117,165}. A local coupling of the surrounding space was done in Ref.¹⁶⁶ by linearizing a non-linear GEP, a similar approach to that of Refs.^{P1,116}.

In this appendix, we propose to quickly re-derive a SAFE method inspired from these previous papers to benchmark the two numerical methods regarding their respective performances. We will use high-order Lobatto function as interpolating function of the finite-elements core of the method. The system studied here is the simple of Lamb modes: an elastic layer of infinite extent in the x_1 direction has two free edges at $x_2 = 0$ and $x_2 = b$, where a no-traction boundary condition is applied.

A.1. Formalism

Let us start with the general variational form for the elasticity equation

$$-\rho\omega^2 \int_{\Omega} u \delta u \, d\Omega + \int_{\Omega} \sigma(u) : \epsilon(\delta u) \, d\Omega = \int_{\partial\Omega} \sigma(\delta u) \cdot \mathbf{n} \, d\Gamma, \quad \forall \delta u, \quad (\text{A.1})$$

where the displacement field is

$$\mathbf{u}(\mathbf{x}) = \mathbf{u}(x_2) e^{ik_1 x_1} e^{-i\omega t}, \quad (\text{A.2})$$

and the double-dot product is $\sigma(u) : \epsilon(\delta u^*) = \sigma_{ij}(u) \epsilon_{ij}(\delta u^*)$. σ designates the stress tensor while ϵ is the strain tensor. Developing this part of the volume term gives the expression,

$$\begin{aligned} \sigma_{11} \epsilon_{11} &= k_1^2 (\lambda + 2\mu) u_1 \delta u_1 - ik_1 \lambda \nabla u_2 \delta u_1 \\ \sigma_{12} \epsilon_{12} &= \frac{\mu}{2} \left(-ik_1 \nabla u_1 \delta u_2 + \nabla u_1 \delta \nabla u_1 + k_1^2 u_2 \delta u_2 + ik_1 u_2 \nabla \delta u_1 \right) \\ \sigma_{22} \epsilon_{22} &= ik_1 \lambda u_1 \nabla \delta u_2 + (\lambda + 2\mu) \nabla u_2 \nabla \delta u_2. \end{aligned} \quad (\text{A.3})$$

Putting these terms back into Eq. A.1 yields,

$$\begin{aligned} \int_{\Omega} -\rho\omega^2 (u_1 \delta u_1 + u_2 \delta u_2) + k_1^2 (\lambda + 2\mu) u_1 \delta u_1 - ik_1 \lambda \nabla u_2 \delta u_1 - ik_1 \mu \nabla u_1 \delta u_2 + \mu \nabla u_1 \delta \nabla u_1 \\ + \mu k_1^2 u_2 \delta u_2 + ik_1 \mu u_2 \nabla \delta u_1 + ik_1 \lambda u_1 \nabla \delta u_2 + (\lambda + 2\mu) \nabla u_2 \nabla \delta u_2 \, d\Omega = 0 \quad \forall \delta u \end{aligned} \quad (\text{A.4})$$

where ∇u_i denotes differentiation of the i -th component of the displacement along x_2 . Note that the surface term initially expressed vanishes from the no-traction boundary condition.

The displacement fields can be discretized as

$$\mathbf{u}(\mathbf{x}) = \mathbf{N}(\xi) \mathbf{U} e^{ik_1 x_1} e^{-i\omega t}, \quad (\text{A.5})$$

with $\mathbf{N}(\xi)$ containing the Lobatto shape functions that spans the local coordinate space $\xi \in [-1, 1]$ of a single element and $\mathbf{U}$ is the vector of displacement amplitudes. Following the

same discretization, the test function thus becomes $\forall \delta U$. By rearranging the terms of Eq. A.4 and writing the weak form for one element,

$$\begin{aligned} \int_{\Omega_e} k_1^2 & \left(\mu \delta U_2^\top M^{(e)} U_2 + (\lambda + 2\mu) \delta U_1^\top M^{(e)} U_1 \right) + i k_1 \left(\mu \delta U_2^\top C_1^{(e)} U - \mu \delta U_1^\top C_2^{(e)} U_2 \right. \\ & \left. + \lambda \delta U_1^\top C_1^{(e)} U_2 - \lambda \delta U_2^\top C_2^{(e)} U_1 \right) + \mu \delta U_1^\top K^{(e)} U_1 + (\lambda + 2\mu) \delta U_2^\top K^{(e)} U_2 \\ & - \rho \omega^2 \delta U_1^\top M^{(e)} U_1 - \rho \omega^2 \delta U_2^\top M^{(e)} U_2 d\Omega_e = 0, \end{aligned} \quad (\text{A.6})$$

where the elementary matrices are,

$$\begin{aligned} M^{(e)} &= \frac{b}{2} \int_{-1}^1 N^\top N d\xi, \quad K^{(e)} = \frac{2}{b} \int_{-1}^1 \nabla N^\top \nabla N d\xi, \\ C_1^{(e)} &= \int_{-1}^1 N^\top \nabla N d\xi, \quad C_2^{(e)} = \int_{-1}^1 \nabla^\top N^\top N U d\xi. \end{aligned} \quad (\text{A.7})$$

In Eq. A.6, the dependance on k_1 was highlighted and terms were sorted depending on their order. The elementary matrices should be assembled into the global matrices. Following an analogous procedure to Chapter. 2, the general form for the problem becomes,

$$\left(k_1^2 A_2 + i k_1 A_1 + A_0 - \omega^2 M \right) U = 0. \quad (\text{A.8})$$

This quadratic generalized eigenvalue problem can be linearized using companion linearization as, ^{tisseur2001, strang2006}

$$\left[\begin{pmatrix} -A_1 & A_0 - \omega^2 M \\ I & 0 \end{pmatrix} - k_1 \begin{pmatrix} A_2 & 0 \\ 0 & I \end{pmatrix} \right] \begin{pmatrix} k_1 U \\ U \end{pmatrix} = 0, \quad (\text{A.9})$$

which solving is thus quite straightforward.

A.2. Results and convergence comparison

Analytical results are obtained by applying the root-finding approach using the Müller method presented in the paper. Solutions are sought from the classical Lamb modes equations, derived in many books such as Royer¹¹⁴. Figure A.1a) shows the results of the three methods for a given frequency range. They all match well. In order to compare the two numerical methods in play, a brief convergence study has been done using the analytical results from the root-finding method as a reference.

The convergence curve of the results given in Fig. A.1b) is unsurprisingly in agreement with the literature. It is calculated at a frequency $f = 4$ kHz: the solutions along the dashed line in Fig A.1a) are selected. The parameter n used for the discretization is shared between the two methods and corresponds to the number of elements taken for the discretization in each method. SAFE matrices have a size $2(2n + 1)$, while SCM matrices have a size $2n$. The discretization is swept from a number of nodes $n = 3$ to $n = 50$. Note that the matrix size for SCM is $4n$ while that of SAFE is $4(o \cdot n + 1)$ with o the order of the Lobatto shape functions.

The SCM convergence shows a steep slope and reaches a plateau around $n = 20$, while the SAFE curve follows the expected slope of a finite elements method that use second-order

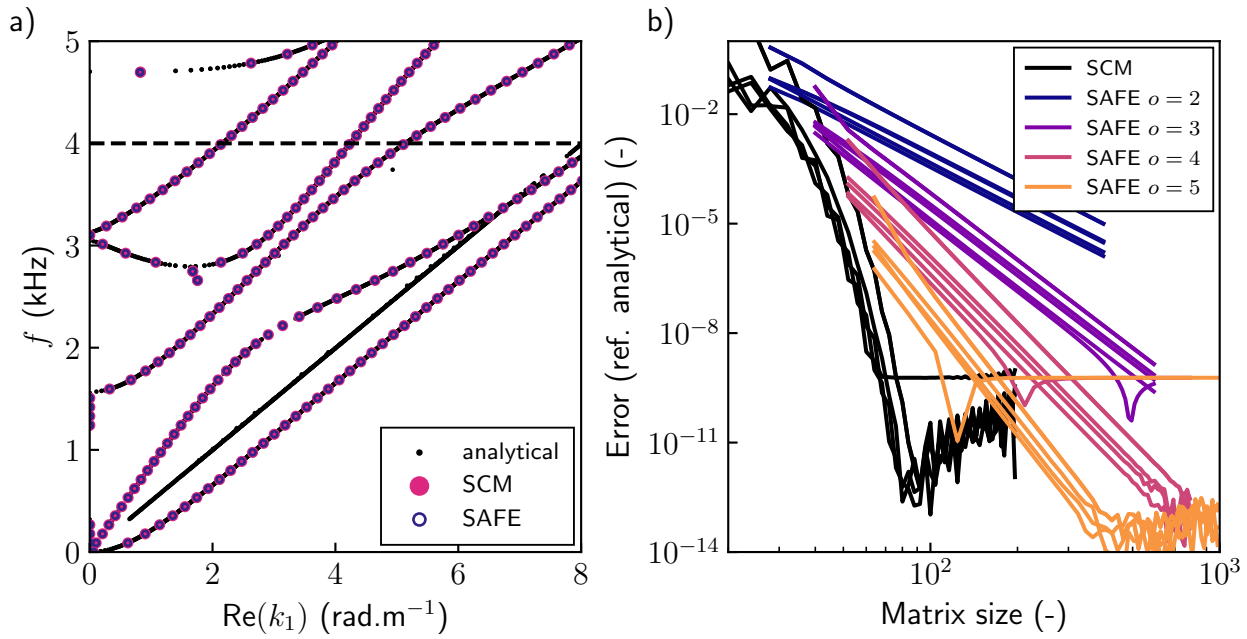


Fig. A.1: a) Dispersion relation for the Lamb modes with the root-finding results, alongside the spectral collocation and semi-analytical finite elements solutions. b) Convergence curve of the 2 methods at a given frequency $f = 4$ kHz. The multiple lines depicts the convergence of each wavenumber.

shape functions. When comparing at a given discretization order, the error from the SAFE is much lower than that of SCM by several orders of magnitude. In order to see comparable convergence, it would require to use higher-order shape functions or use spectral elements instead.

B | Details on the root-finding algorithm

In this appendix, we will give a few explanations on the root-finding procedure that has been developed throughout this thesis. Its core is based on the Müller root-finding method, but several procedures are used around in order to make the procedure efficient and reliable.

B.1. The Müller method

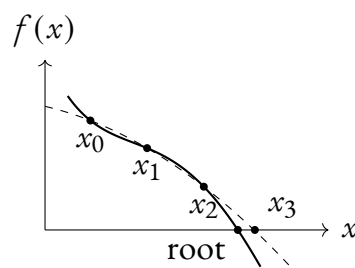


Fig. B.1: Illustration of the Müller iterative scheme: the root of function $f(x)$ (—) is approximated with an interpolation function (---) by the x_3 point.

B. Details on the root-finding algorithm

This method is a root-finding procedure that relies on a quadratic interpolation of the function, here denoted as f . From three initial guesses close to each other that are x_0, x_1 and x_2 , a parabola is constructed by interpolating the function of interest at the three initial guesses.^{134,167} The parabola $p(x)$ has an expression written as $p(x_3) = a(x_3 - x_2)^2 + b(x_3 - x_2) + c$ and, denoting $f_i = f(x_i)$, passes through the points (x_0, f_0) , (x_1, f_1) and (x_2, f_2) . The coefficients for this quadratic polynomial are thus given as,

$$\begin{aligned} a &= \frac{(x_1 - x_2)(f_0 - f_2) - (x_0 - x_2)(f_1 - f_2)}{\prod_{0 \leq i < j \leq 2} (x_j - x_i)}, \\ b &= \frac{(x_0 - x_2)^2(f_1 - f_2) - (x_1 - x_2)^2(f_0 - f_2)}{\prod_{0 \leq i < j \leq 2} (x_j - x_i)}, \\ c &= f_2. \end{aligned} \tag{B.1}$$

The root of this polynomial is thus expressed as,

$$x_3 = x_2 - \frac{2c}{b + \text{sign}(b)\sqrt{b^2 - 4ac}}, \tag{B.2}$$

and chosen such that the denominator should be the biggest value, hence the sign function. While not the classically written quadratic formula, this expression remains equivalent and allows to avoid round-off errors for small values of a ¹⁶⁸. This process is then run iteratively until the condition for finding the root is met,

$$\varepsilon_a = \left| \frac{x_3 - x_2}{x_3} \right|, \tag{B.3}$$

which means that the convergence is reached when the iteratively-found solution stops varying up to small values; we have usually chosen $\varepsilon_a \in [10^{-12}, 10^{-18}]$ depending on the nature of the problem and more importantly the numerical conditioning of the complex function f . It is possible to initialize the method with initial guesses that leads to a divergence of the method, or to location which never converge. To mitigate these two issues, a second criterion $\varepsilon_b \approx 10^6$ sets a threshold for the variance of $[x_0, x_1, x_2, x_3]$ for the first case, and to a parameter setting the maximum number of iterations set in the range $[100, 2000]$.

B.2. A complete routine for the automated solving of dispersion relations

In this section is described the algorithm that was developed around the Müller method for the automatic computation of dispersion relations. A diagram is presented in Fig. B.2. The method is initialized with all necessary parameters, among which an array of frequencies and the function of interest. The first iteration usually does not contain any initial guesses, so a mapping of the function is done which depends on the range of values given as the input. An example of a map is shown in the diagram, from which each local minimum k_g is easily found and represented as the black dots, corresponding to the lowest map values. Each of these

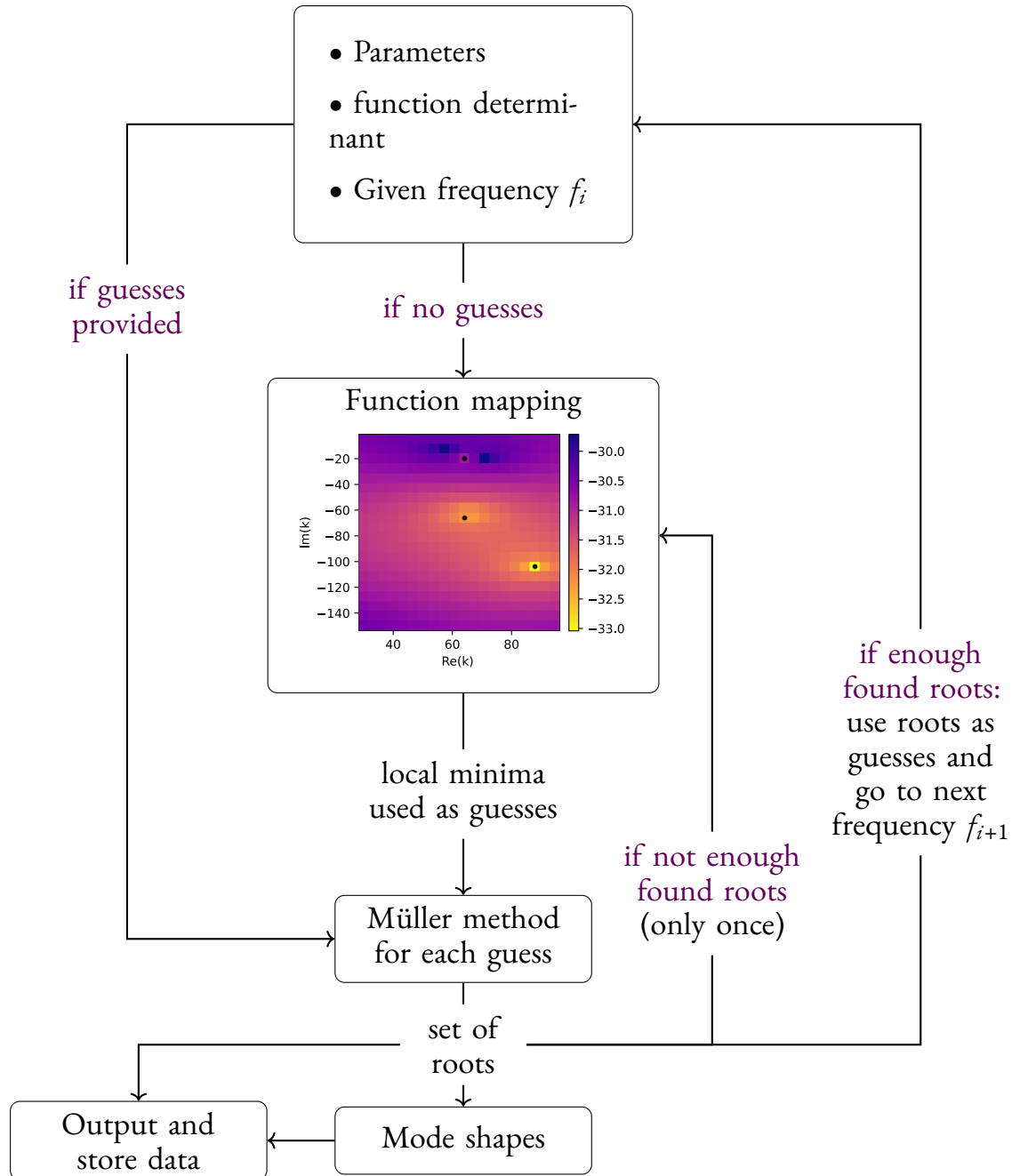


Fig. B.2: Description of the routine. Accented text corresponds to conditions and text not embedded in a box corresponds to the data obtained in the previous process. The map is shown in logarithmic scale of the absolute value of the function value depending on the complex coordinate of the wavenumber.

solutions leads to a guess used for the initialisation of the method, which are given as $k_g - \delta k$, k_g and $k_g + \delta k$, with δk and arbitrary value set as $\delta k = 1 + i \cdot 1$ for most cases. It was observed that the variation of this parameter does not affect the results as long as it is kept to rather small values of the magnitude of the solution to solve.

Next, the Müller method is run for each of this set of parameters. This yields a set of converged roots. If the number of converged roots is the one expected from the input value of the procedure, then the program goes to the next iteration. Otherwise, a refined map of the function is evaluated and the iterative process is run again. Roots are checked to lie in the input range for the wavenumber and that they are all unique, since converging from multiple sets of guesses does not require that the method will not converge several times to the same root; this is checked with another threshold $\varepsilon_c \approx 10^{-8}$. Once enough unique roots have been found, the latter are fed to the procedure as the new initial guesses: if the frequency change is small enough, then the roots should be close to the previous ones as each branch of a dispersion relation should be continuous in most of the frequency spectrum. If enough roots have been provided as initial guesses, the program avoids computing a mapping of the function, thus saving an rather important computation time.

Once the set of solutions has been finally obtained, each of the associated mode shapes can also be retrieved. Given a specific root $k_{1,j}$ that satisfies $\det(\mathbf{M}) = 0$, there is $\mathbf{M}(k_{1,j})\mathbf{S} = 0$, where the vector $\mathbf{S} = \ker(\mathbf{M}(k_{1,j}))$ is the nullspace of the matrix. In practice, the nullspace is computed using singular value decomposition (SVD), $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\dagger$. The last right singular vectors, that is the last row of $\mathbf{V}^\dagger$ provide an orthonormal basis for the null space of $\mathbf{M}$, associated to the singular value $\sigma = \min(\mathbf{\Sigma})$.

B.3. Using the code

In order to use the developed method, one should write a Python class formulating the problem, the requirements are that there should exist an `update_frequency` method taking an angular frequency as the only input and should update all frequency-dependant parameters. Also, there should exist a `get_matrix` method taking a complex wavenumber (or any parameter that needs to be root-found) and returns either a matrix or a scalar. The code is available on Github [déposer sur zenodo](#) and more practical informations are given there.

C | Overview of activities and scientific works during the thesis

C.1. Publications

- P1. M. Maréchal, A. Geslain, J.-P. Groby, V. Romero-García, and O. Dazel. “A General Spectral Collocation Method for Computing the Dispersion Relations of Guided Acoustic Waves in Multilayer Dissipative Structures”. *J. App. Phys.* 137.10 (2025), 104902.
- P2. M. Maréchal, O. Dazel, V. Romero-García, and J.-P. Groby. “Numerical Models for the Scattering Properties of Multilayered Structures with Periodic Cells”. *submitted to Int. J. Numer. Meth. Eng.* (2025).

C.2. Conferences

(ajouter titres des présentations)

- SAM 2022 Nimes, France, October 2022 (attendee)
- JJCAB Toulouse 2023, France, July 2023 (poster)
- SAPEM Sorrento, Italy, October 2023 (presenter)
- Séminaire de 2e année Centrale Nantes, France, April 2024 (presenter)
- SAM Aegina, Greece, May 2024 (presenter)
- INTERNOISE Nantes, France, August 2024 (attendee and volunteer)
- CFA 2025 Paris, France (presenter)

C.3. Workshops and training schools

- training school Acoustic Metamaterials Valencia Spain, UPV, November 2023
- lamb waves workshop

C.4. Teaching activities

D | Extended abstract in french

Dans cette thèse nous nous intéressons aux structures métaporeuses, en particulier à leur comportement quant à la diffusion par ces structures d'une onde plane incidente ainsi qu'à leur réponse libre permettant de calculer les modes complexes de ces structures.

En INTRODUCTION, le contexte de cette étude lié à l'impact généralisé du bruit dans nos sociétés post-industrielles est présenté. Nous nous intéressons à l'amélioration des performances d'isolation et d'absorption acoustique par l'utilisation d'un certain type de métamatériau acoustique qu'on appelle les surfaces et les interfaces métaporeuses. Un état de l'art détaillé est réalisé à ce sujet, détaillant succinctement l'historique de ce champ d'étude. Une partie est consacrée aux métamatériaux acoustiques au sens large. Une dernière s'intéresse à l'utilisation de matériaux poreux et poroélastiques pour développer des métamatériaux.

Le CHAPITRE 1 présente le cadre théorique nécessaire à la compréhension des travaux effectués dans la thèse, reprenant diverses méthodes permettant le calcul de la réponse acoustique de structures.

Dans le CHAPITRE 2, l'intérêt est consacré aux calcul des propriétés de diffusion par des multicouches composés de divers types de matériaux, incluant des inclusions périodiques. Un formalisme est développé à cette fin, permettant le couplage d'une méthode éléments finis pour la modélisation des coeurs périodiques avec une modélisation par ondes planes des couches homogènes. Ce couplage est effectué par l'expression d'un vecteur d'état à chacune des interfaces du système. Par la suite, la convergence de la méthode est montrée en comparaison avec d'autres méthodes existantes (TMM). La stabilité et la robustesse de la présente méthode est montrée par l'étude des coefficients de diffusions de structures déjà étudiées auparavant dans la littérature.

Le calcul numérique de relations de dispersion d'une structure multicouche arbitraire est l'objet du CHAPITRE 3. Une méthode numérique intitulé SCM est utilisé et implémenté pour un formalisme général permettant la modélisation d'ondes guidées dans des multicouches illimités dans la direction longitudinale. A chaque frontière du système dans la direction transverse, le rayonnement des ondes guidées est possible en dehors de la structure, dans un demi-espace fluide. L'investigation de résultats est faite pour le cas d'un bicouche élastique-poroélastique avec du rayonnement dans l'air. La relation de dispersion des modes complexes et évanescents est présentée pour cette configuration. Une comparaison par des résultats expérimentaux obtenus à l'aide de la méthode SLaTCoW est aussi abordé.

Dans le CHAPITRE 4, l'objet d'étude est la relation de dispersion dans des couches périodiques poreuses.

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